



**NTNU – Trondheim**  
Norwegian University of  
Science and Technology

# **TET4185: Power Markets, Resources and Environment**

## **Grid-Constrained Market Clearing**

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# Outline

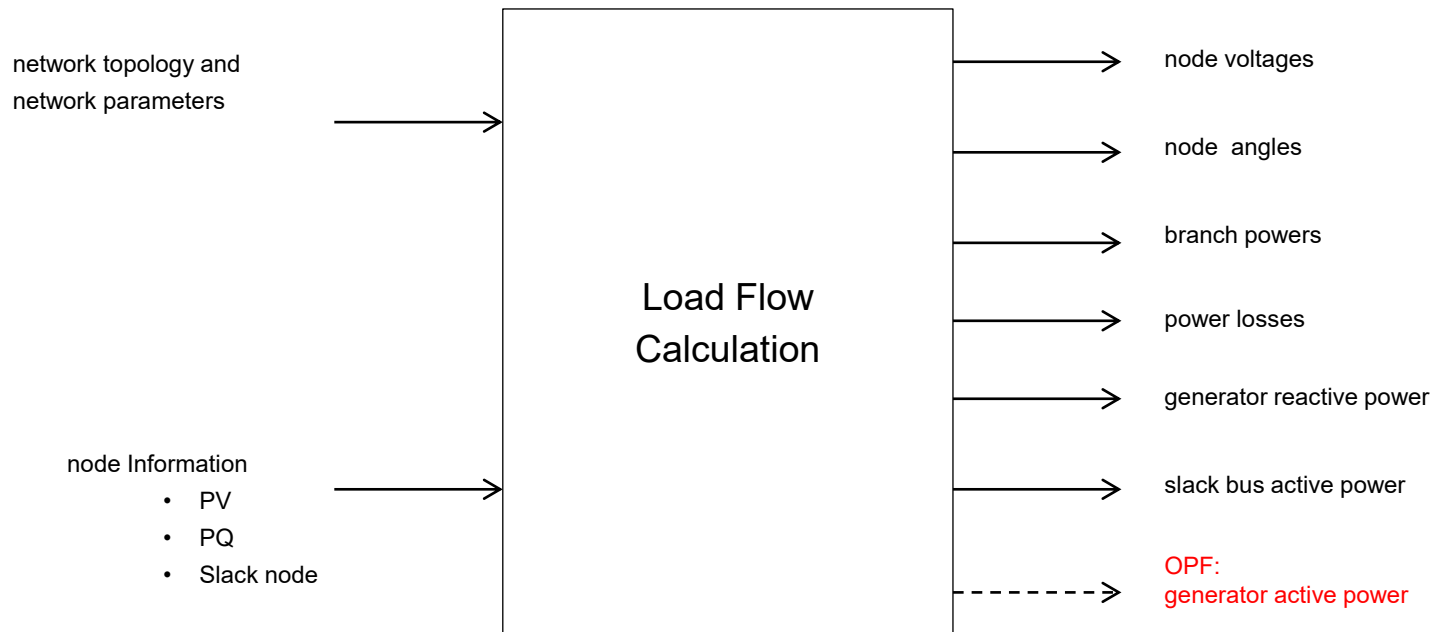
- DC Load Flow Foundations
- Grid-Constrained Market Clearing (ATC vs DC-OPF)
- PTDF and Flow-Based Representation
- Price Formation and Market Coupling

# DC Load flow

- Ref Memo "DC Power Flow in Power Markets"
- First part (pages 3-10) is background
- Second part (page 11), from "DC load flow" is curriculum
- Can be seen as an extension of Section 7.3.9 in the textbook
- Section 7.3.10 ("Hydro and hydro-thermal systems") in the book is not in the curriculum



# Load Flow – general approach (very short summary)



# Admittance matrix

$$\mathbf{I} = \mathbf{YV} \Leftrightarrow \begin{bmatrix} I_1 \\ I_2 \\ \cdot \\ \cdot \\ I_N \end{bmatrix} = \begin{bmatrix} \color{red}{Y_{11}} & Y_{12} & \cdot & \cdot & Y_{1N} \\ Y_{21} & \color{red}{Y_{22}} & \cdot & \cdot & Y_{2N} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ Y_{N1} & Y_{N2} & \cdot & \cdot & \color{red}{Y_{NN}} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ \cdot \\ \cdot \\ V_N \end{bmatrix}$$

- The element on the main diagonal  $Y_{ii}$  equals the sum of all admittances directly connected to node  $i$
- The off-diagonal element  $Y_{ij}$  equals the negative value of the net admittance connected between node  $i$  and node  $j$

(note that the off-diagonal element  $\underline{Y}_{ji}$  has the same value as  $\underline{Y}_{ij}$ .)

$$Z = \frac{V}{I} = R + jX \Rightarrow Y = \frac{I}{V} = G + jB$$

$$G = \frac{R}{R^2 + X^2} \quad \text{and} \quad B = -\frac{X}{R^2 + X^2}$$

Z: Impedance

R: Resistance

X: Reactance

Y: Admittance

G: Conductance

B: Susceptance

# Power Equations- Complex Power

$$\begin{aligned}
 S_i &= VI^* = |V_i| \angle \delta_i \sum_{n=1}^N Y_{in}^* V_n^* = \sum_{n=1}^N |V_i V_n Y_{in}| \angle (-\theta_{in} - \delta_n + \delta_i) \\
 &= \sum_{n=1}^N |V_i V_n Y_{in}| [\cos(-\theta_{in} - \delta_n + \delta_i) + j \sin(-\theta_{in} - \delta_n + \delta_i)] \\
 &= \sum_{n=1}^N |V_i V_n Y_{in}| [\cos(\theta_{in} + \delta_n - \delta_i) - j \sin(\theta_{in} + \delta_n - \delta_i)]
 \end{aligned}$$

$I^*$  is the complex conjugate of  $I$

$$\begin{aligned}
 P_i &= \sum_{n=1}^N |V_i V_n Y_{in}| \cos(\theta_{in} + \delta_n - \delta_i) & Q_i &= -\sum_{n=1}^N |V_i V_n Y_{in}| \sin(\theta_{in} + \delta_n - \delta_i) \\
 &= |V_i|^2 G_{ii} + \sum_{\substack{n=1 \\ n \neq i}}^N |V_i V_n Y_{in}| \cos(\theta_{in} + \delta_n - \delta_i) & &= -|V_i|^2 B_{ii} - \sum_{\substack{n=1 \\ n \neq i}}^N |V_i V_n Y_{in}| \sin(\theta_{in} + \delta_n - \delta_i)
 \end{aligned}$$

These equations are solved in an iterative scheme, e.g., the so-called Newton-Raphson

$$\left. \begin{array}{l} g(x) = 0 \\ g(x^0 + \Delta x^0) = 0 \end{array} \right\} \xrightarrow{\text{Taylor series}} g(x^0 + \Delta x^0) = g(x^0) + g_x(x^0) \Delta x^0 = 0 \Rightarrow \Delta x^0 = -\frac{g(x^0)}{g_x(x^0)} \Rightarrow x^{k+1} = x^k - \frac{g(x^k)}{g_x(x^k)}$$

# Jacobian Matrix

- The **Jacobian matrix** is the matrix of all first-order partial derivatives of a vector-valued function with respect to another vector.

$$F = \begin{bmatrix} F_1 \\ \vdots \\ F_m \end{bmatrix}, X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \Rightarrow J = \frac{dF}{dX} = \begin{bmatrix} \frac{\partial F_1}{\partial X_1} & \cdots & \frac{\partial F_1}{\partial X_n} \\ \vdots & & \vdots \\ \frac{\partial F_m}{\partial X_1} & \cdots & \frac{\partial F_m}{\partial X_n} \end{bmatrix}$$

- If the function **F** is differentiable at a point **x<sub>0</sub>** then the Jacobian matrix is the best linear approximation of the function **F** near the point **x<sub>0</sub>** (Taylor series)
- Named after the mathematician Carl Gustav Jacob Jacobi (1804-1851)



# Jacobi Matrix

The first-order derivatives of the power mismatches w.r.t. the voltage angles and the voltage magnitudes

$$\mathbf{J}\Delta\mathbf{x} = \begin{bmatrix} \Delta\mathbf{P}(\mathbf{x}) \\ \Delta\mathbf{Q}(\mathbf{x}) \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} \boldsymbol{\delta} \\ |\mathbf{V}| \end{bmatrix} = \begin{bmatrix} \delta_2 \\ \vdots \\ \delta_N \\ |V|_2 \\ \vdots \\ |V|_N \end{bmatrix}$$

$$\mathbf{J} = \begin{bmatrix} \mathbf{J}_{11} & \mathbf{J}_{12} \\ \mathbf{J}_{21} & \mathbf{J}_{22} \end{bmatrix}$$

$$J_{11} \quad \text{elements of type } \frac{\partial P}{\partial \delta}$$

$$J_{12} \quad \text{elements of type } \frac{\partial P}{\partial V}$$

$$J_{21} \quad \text{elements of type } \frac{\partial Q}{\partial \delta}$$

$$J_{22} \quad \text{elements of type } \frac{\partial Q}{\partial V}$$

Note:  $\mathbf{J}$  must be re-established for each iteration (can be seen by writing the expressions of the partial derivatives)



# Decoupled load flow-1

- Overhead transmission lines:  $R_{ij} \ll X_{ij}$
- The differences between the voltage angles normally small
- $\mathbf{J}_{12}$  and  $\mathbf{J}_{21}$  much smaller than  $\mathbf{J}_{11}$  and  $\mathbf{J}_{22}$
- $$\left. \begin{array}{l} \mathbf{J}_{11} \Delta \delta = \Delta \mathbf{P} \\ \mathbf{J}_{22} \Delta |\mathbf{V}| = \Delta \mathbf{Q} \end{array} \right\} \quad \mathbf{J}_{11} \text{ and } \mathbf{J}_{22} \text{ have to be recalculated}$$
- Further simplifications:
  - $G_{ij} \sin(\delta_j - \delta_i) \ll B_{ij} \cos(\delta_j - \delta_i)$
  - $\sin(\delta_j - \delta_i) \approx \delta_j - \delta_i$       and       $\cos(\delta_j - \delta_i) \approx 1$
  - $Q_i \ll |V_i|^2 B_{ii}$

# Decoupled load flow-2

$$\begin{bmatrix} -B_{22} & \cdot & \cdot & -B_{2N} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ -B_{N2} & \cdot & \cdot & -B_{NN} \end{bmatrix} \begin{bmatrix} \Delta\delta_2 \\ \cdot \\ \cdot \\ \Delta\delta_N \end{bmatrix} = \begin{bmatrix} \Delta P_2/|V_2| \\ \cdot \\ \cdot \\ \Delta P_N/|V_N| \end{bmatrix} \quad \begin{bmatrix} -B_{22} & \cdot & \cdot & -B_{2N} \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ -B_{N2} & \cdot & \cdot & -B_{NN} \end{bmatrix} \begin{bmatrix} \Delta|V_2| \\ \cdot \\ \cdot \\ \Delta|V_N| \end{bmatrix} = \begin{bmatrix} \Delta Q_2/\Delta|V_2| \\ \cdot \\ \cdot \\ \Delta Q_N/\Delta|V_N| \end{bmatrix}$$

- $B_{ij}$  are line susceptances
- This means that the elements of the Jacobi matrix can be calculated once instead of for each calculations
- Significantly speeds up calculation time per iteration
  - at the cost of less accuracy and therefore more iterations
- Normally faster
- Decoupled load flow is AC load flow, but faster algorithm

# DC load flow

- We only focus on active power and voltage angles
- Set all node voltages = 1
- Neglect all resistances, i.e.  $Y_{ij} = B_{ij}$
- Differences between voltage angles are small
  - $\sin(\delta_j - \delta_i) \approx \delta_j - \delta_i$  and  $\cos(\delta_j - \delta_i) \approx 1$

- Can then write 
$$P_i \approx \sum_{\substack{n=1 \\ n \neq i}}^N |B_{in}| (\delta_i - \delta_n)$$
- Line flows given by 
$$P_{ij} = \frac{1}{x_{ij}} (\delta_i - \delta_j)$$
- Important: load flow problem becomes linear problem
  - no need for iteration procedure!

# Grid-Constrained Market Clearing (ATC vs DC-OPF)

# Market clearing problem with inelastic load-optimal dispatch problem (matrix formula)

$$\begin{aligned} & \min_P f(\mathbf{P}) \\ & \text{subject to :} \\ & \quad \sum \mathbf{P} = \mathbf{D} \\ & \quad \mathbf{P}_{\min} \leq \mathbf{P} \leq \mathbf{P}_{\max} \end{aligned}$$

$f$  : The cost function of all generators

$\mathbf{P}$  : Vector of actual generation

$\mathbf{D}$  : The system demand

$\mathbf{P}_{imin} , \mathbf{P}_{imax}$  : The vector of minimum and maximum production respectively

# Market clearing problem with grid constraints – ATC Approach

$$\min_{P, FL} f(P)$$

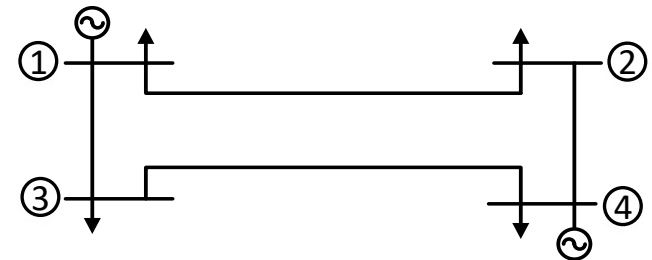
subject to :

$$P - D = I \times FL$$

(one constraint for each node)

$$FL_{\min} \leq FL \leq FL_{\max}$$

$$P_{\min} \leq P \leq P_{\max}$$



- **FL** are the line flows
  - **I** is the bus-branch incidence matrix,  $[N_{\text{nodes}} \times N_{\text{lines}}]$ 
    - $I(i,j)$  is 1 if  $i$  is the *from* node of the line
    - $I(i,j)$  is -1 if  $i$  is the *to* node of the line
    - $I(i,j)$  is 0 otherwise
  - The first constraint corresponds to  $g(U, P) = 0$
  - The second constraint corresponds to  $h(U) = 0$
  - The third constraint corresponds to  $P - P_{\max} \leq 0$
- } **DC grid**

# Market clearing problem as DC-OPF

$$\min_{P, \Delta} f(P)$$

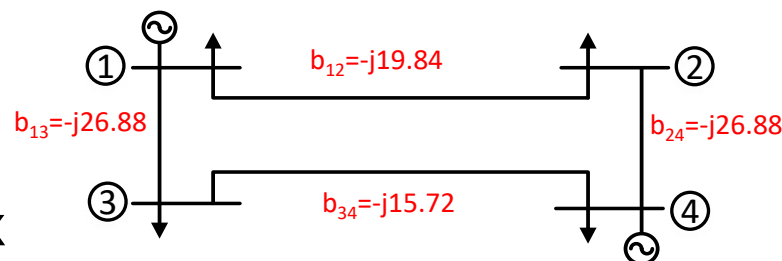
subject to :

$$P - D = \overset{B}{jY} \times \Delta$$

(one constraint for each node)

$$FL_{\min} \leq F\Delta \leq FL_{\max}$$

$$P_{\min} \leq P \leq P_{\max}$$



- $Y$  is the bus admittance matrix
- $\Delta$  are the bus angles in radians
- $F$  is a connection matrix (branch-bus incidence matrix)  $[N_{\text{lines}} \times N_{\text{nodes}}]$  matrix
  - $F(i,j)$  is  $|b_{ij}|$  if  $i$  is the from node of the line
  - $F(i,j)$  is  $-|b_{ij}|$  if  $i$  is the to node of the line
  - $F(i,j)$  is 0 otherwise
- The first constraint corresponds to  $g(U,P) = 0$
- etc

} DC grid

## ATC Approach (commercial exchanges)

$$\min_{P, FL} f(\mathbf{P})$$

$$\begin{aligned} \text{subject to :} \quad & \mathbf{P} - \mathbf{D} = \mathbf{I} \times \mathbf{FL} \\ & \mathbf{FL}_{\min} \leq \mathbf{FL} \leq \mathbf{FL}_{\max} \\ & \mathbf{P}_{\min} \leq \mathbf{P} \leq \mathbf{P}_{\max} \end{aligned}$$

## OPF problem (Physical exchanges)

$$\min_{P, \Delta} f(\mathbf{P})$$

$$\begin{aligned} \text{subject to :} \quad & \mathbf{P} - \mathbf{D} = \mathbf{B} \times \Delta \\ & \mathbf{FL}_{\min} \leq \mathbf{F} \Delta \leq \mathbf{FL}_{\max} \\ & \mathbf{P}_{\min} \leq \mathbf{P} \leq \mathbf{P}_{\max} \end{aligned}$$

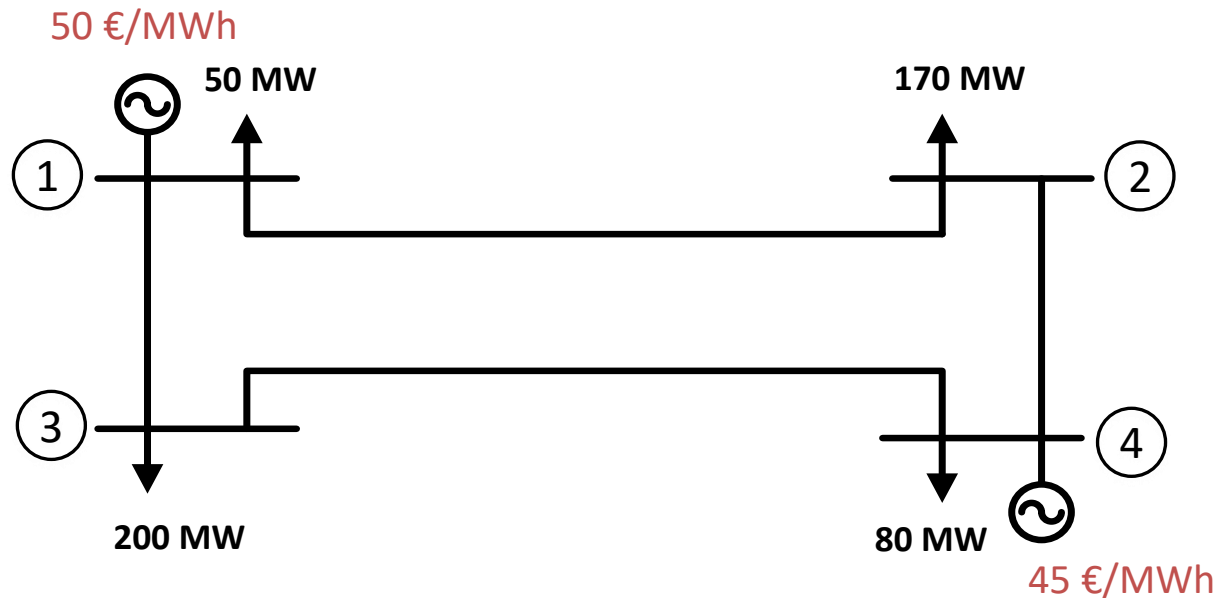
### Physical limitations in the grid must be respected during operation:

- Imposed as limits to commercial exchanges (the electricity market), or
- Managed directly (by counter trade or re-dispatch during operation)





# OPF example grid



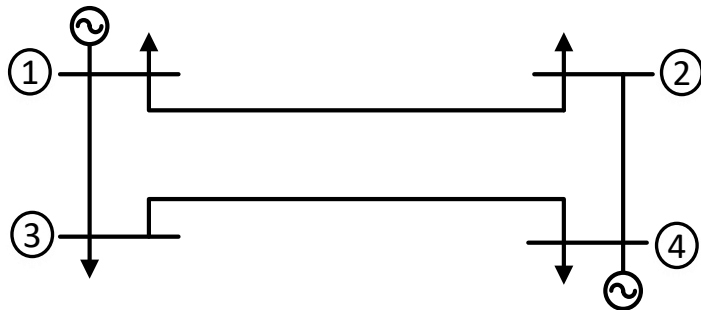
| line | from-to | B (per unit) |
|------|---------|--------------|
| 1    | 1-2     | -j19.84      |
| 2    | 1-3     | -j26.88      |
| 3    | 2-4     | -j26.88      |
| 4    | 3-4     | -j15.72      |

Per unit base:

$S_N = 100$  MVA

$V_N = 230$  kV

# Dispatch problem with grid constraints – ATC Approach



Line flows

$$\begin{bmatrix} FL_{12} \\ FL_{13} \\ FL_{24} \\ FL_{34} \end{bmatrix}$$

Bus-branch incident matrix  $[N_{\text{nodes}} \times N_{\text{lines}}]$

$$I = \begin{matrix} & \begin{matrix} FL_{12} & FL_{13} & FL_{24} & FL_{34} \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \end{matrix}$$

and

$$P_1 - D_1 = FL_{12} + FL_{13}$$

$$P_2 - D_2 = -FL_{12} + FL_{24}$$

$$P_3 - D_3 = -FL_{13} + FL_{34}$$

$$P_4 - D_4 = -FL_{24} - FL_{34}$$

$$\Rightarrow \begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix} - \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} FL_{12} \\ FL_{13} \\ FL_{24} \\ FL_{34} \end{bmatrix}$$

# Dispatch problem with grid constraints – ATC Approach

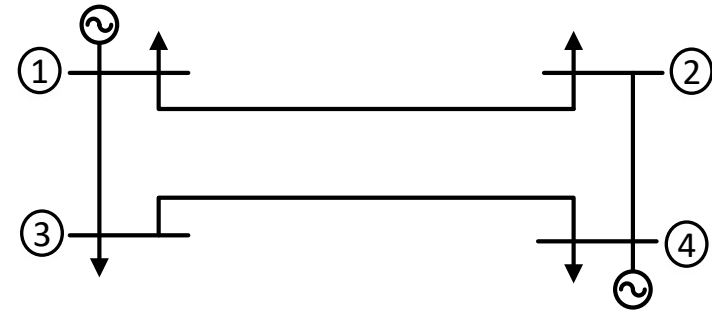
$$\min_{P, FL} (c_1 P_1 + c_4 P_4)$$

subject to :

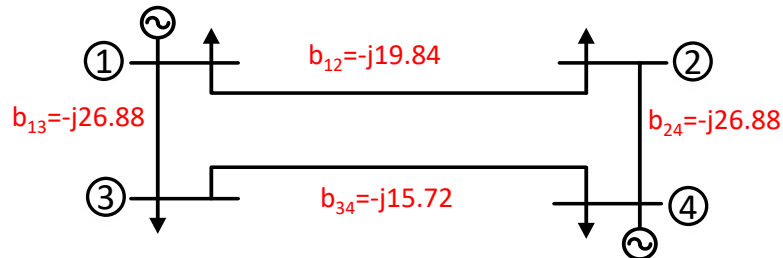
$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix} - \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} FL_{12} \\ FL_{13} \\ FL_{24} \\ FL_{34} \end{bmatrix}$$

$$\begin{bmatrix} FL_{12\min} \\ FL_{13\min} \\ FL_{24\min} \\ FL_{34\min} \end{bmatrix} \leq \begin{bmatrix} FL_{12} \\ FL_{13} \\ FL_{24} \\ FL_{34} \end{bmatrix} \leq \begin{bmatrix} FL_{12\max} \\ FL_{13\max} \\ FL_{24\max} \\ FL_{34\max} \end{bmatrix}$$

$$\begin{bmatrix} P_{1\min} \\ P_{2\min} \end{bmatrix} \leq \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} \leq \begin{bmatrix} P_{1\max} \\ P_{2\max} \end{bmatrix}$$



# OPF problem with constraints



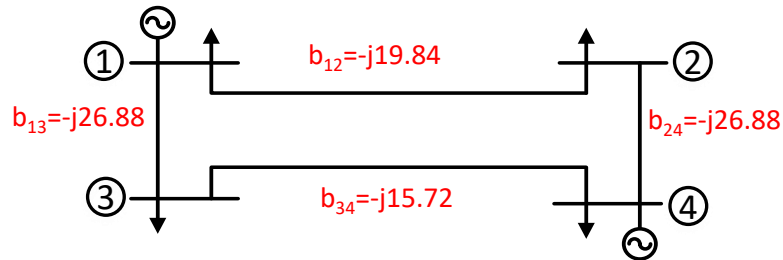
Admittance matrix

$$\mathbf{Y} = j \begin{bmatrix} -46.72 & 19.84 & 26.88 & 0 \\ 19.84 & -46.72 & 0 & 26.88 \\ 26.88 & 0 & -42.6 & 15.72 \\ 0 & 26.88 & 15.72 & -42.6 \end{bmatrix} \Rightarrow \mathbf{B} = j\mathbf{Y} = \begin{bmatrix} 46.72 & -19.84 & -26.88 & 0 \\ -19.84 & 46.72 & 0 & -26.88 \\ -26.88 & 0 & 42.6 & -15.72 \\ 0 & -26.88 & -15.72 & 42.6 \end{bmatrix}$$

Connection matrix  $[\mathbf{N}_{\text{lines}} \times \mathbf{N}_{\text{nodes}}]$

$$\mathbf{F} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} L_{12} \\ L_{13} \\ L_{24} \\ L_{34} \end{matrix} & \begin{bmatrix} 19.84 & -19.84 & 0 & 0 \\ 26.88 & 0 & -26.88 & 0 \\ 0 & 26.88 & 0 & -26.88 \\ 0 & 0 & 15.72 & -15.72 \end{bmatrix} \end{matrix}$$

# OPF problem with constraints



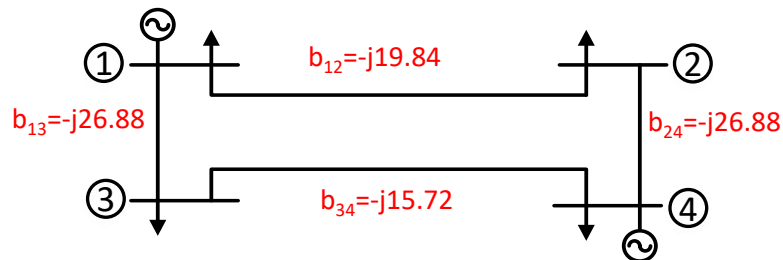
## Power Flow equations

$$\begin{bmatrix} P_1 \\ P_2 \\ P_3 \\ P_4 \end{bmatrix} - \begin{bmatrix} D_1 \\ D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 46.72 & -19.84 & -26.88 & 0 \\ -19.84 & 46.72 & 0 & -26.88 \\ -26.88 & 0 & 42.6 & -15.72 \\ 0 & -26.88 & -15.72 & 42.6 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$

bus 1 is chosen as the reference bus  $\delta_1 = 0$

$$\begin{bmatrix} P_2 \\ P_3 \\ P_4 \end{bmatrix} - \begin{bmatrix} D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 46.72 & 0 & -26.88 \\ 0 & 42.60 & -15.72 \\ -26.88 & -15.72 & 42.60 \end{bmatrix} \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$

# OPF problem with constraints



$$\mathbf{F} = \begin{bmatrix} -19.84 & 0 & 0 \\ 0 & -26.88 & 0 \\ 26.88 & 0 & -26.88 \\ 0 & 15.72 & -15.72 \end{bmatrix}$$

## Transmission constraints

$$\begin{bmatrix} FL_{12 \min} \\ FL_{13 \min} \\ FL_{24 \min} \\ FL_{34 \min} \end{bmatrix} \leq \begin{bmatrix} 19.84 & -19.84 & 0 & 0 \\ 26.88 & 0 & -26.88 & 0 \\ 0 & 26.88 & 0 & -26.88 \\ 0 & 0 & 15.72 & -15.72 \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} \leq \begin{bmatrix} FL_{12 \max} \\ FL_{13 \max} \\ FL_{24 \max} \\ FL_{34 \max} \end{bmatrix}$$

$$\begin{bmatrix} FL_{12 \min} \\ FL_{13 \min} \\ FL_{24 \min} \\ FL_{34 \min} \end{bmatrix} \leq \begin{bmatrix} -19.84 & 0 & 0 \\ 0 & -26.88 & 0 \\ 26.88 & 0 & -26.88 \\ 0 & 15.72 & -15.72 \end{bmatrix} \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} \leq \begin{bmatrix} FL_{12 \max} \\ FL_{13 \max} \\ FL_{24 \max} \\ FL_{34 \max} \end{bmatrix}$$

# OPF problem with constraints

$$\min_{P, \Delta} (c_1 P_1 + c_4 P_4)$$

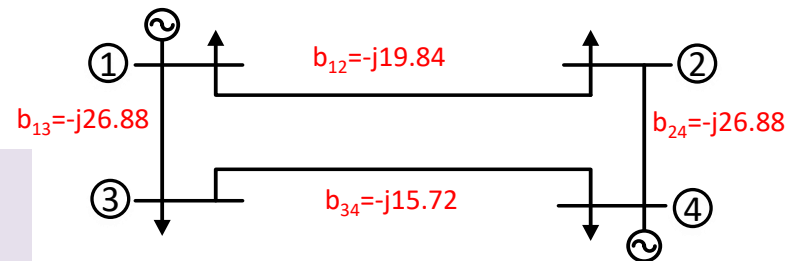
subject to :

$$P_1 = D_1 + \sum_{i=2}^4 (D_i - P_i)$$

$$\begin{bmatrix} P_2 \\ P_3 \\ P_4 \end{bmatrix} - \begin{bmatrix} D_2 \\ D_3 \\ D_4 \end{bmatrix} = \begin{bmatrix} 46.72 & 0 & -26.88 \\ 0 & 42.60 & -15.72 \\ -26.88 & -15.72 & 42.60 \end{bmatrix} \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix}$$

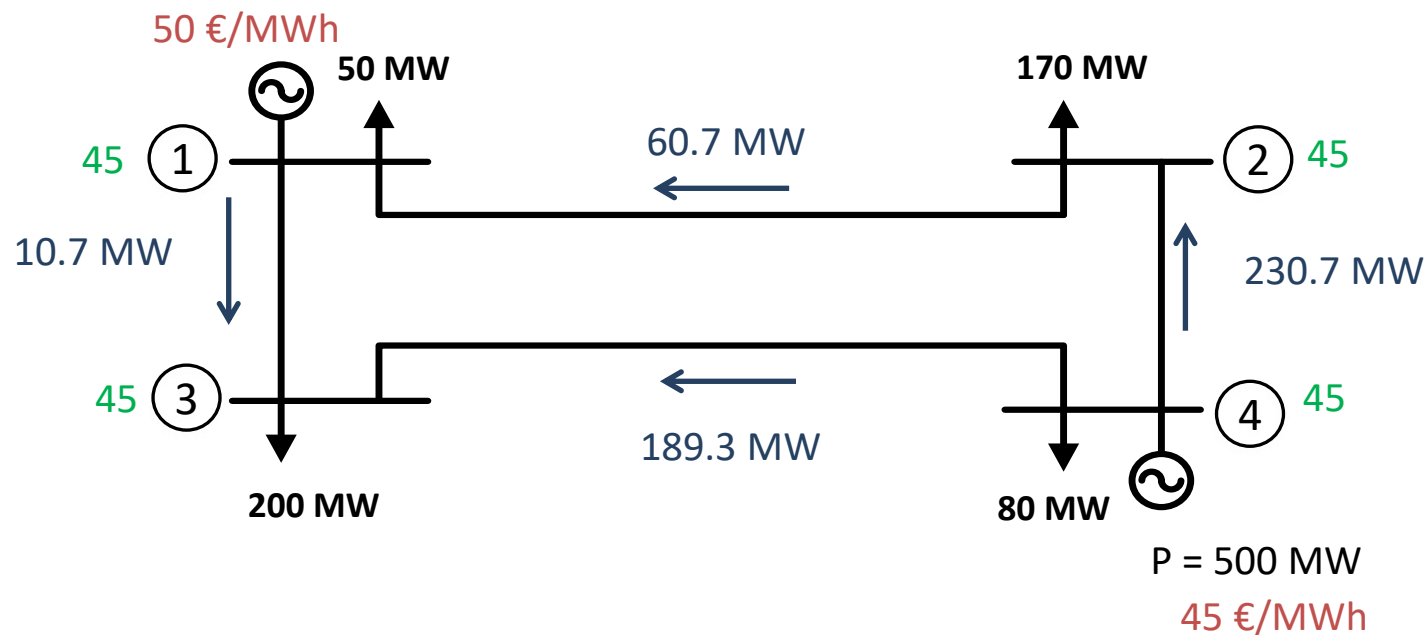
$$\begin{bmatrix} FL_{12\min} \\ FL_{13\min} \\ FL_{24\min} \\ FL_{34\min} \end{bmatrix} \leq \begin{bmatrix} -19.84 & 0 & 0 \\ 0 & -26.88 & 0 \\ 26.88 & 0 & -26.88 \\ 0 & 15.72 & -15.72 \end{bmatrix} \begin{bmatrix} \delta_2 \\ \delta_3 \\ \delta_4 \end{bmatrix} \leq \begin{bmatrix} FL_{12\max} \\ FL_{13\max} \\ FL_{24\max} \\ FL_{34\max} \end{bmatrix}$$

$$\begin{bmatrix} P_{1\min} \\ P_{2\min} \end{bmatrix} \leq \begin{bmatrix} P_1 \\ P_2 \end{bmatrix} \leq \begin{bmatrix} P_{1\max} \\ P_{2\max} \end{bmatrix}$$



# Copper plate solution and real flow

- $P_{G4} = 500 \text{ MW}$ , cost = 22500 €/hour green: prices

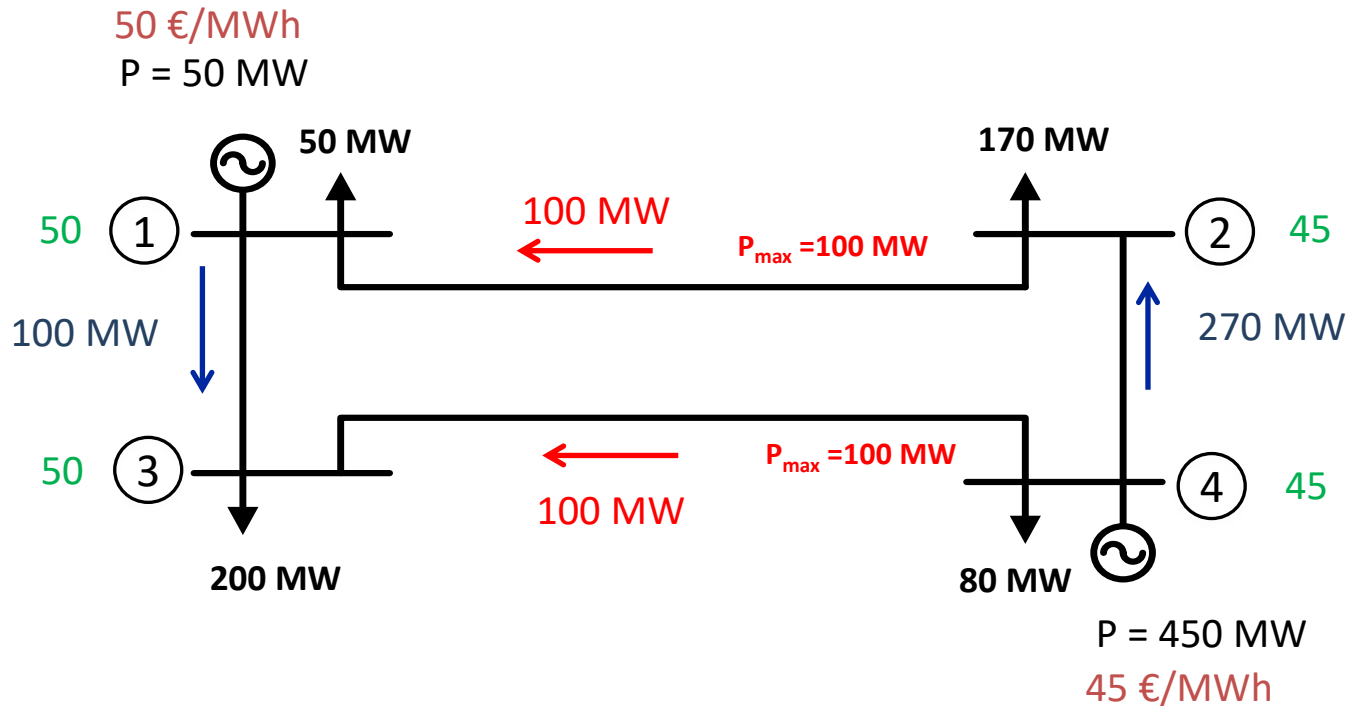




# ATC model with transmission constraints

- Cost = 22750 €/hour

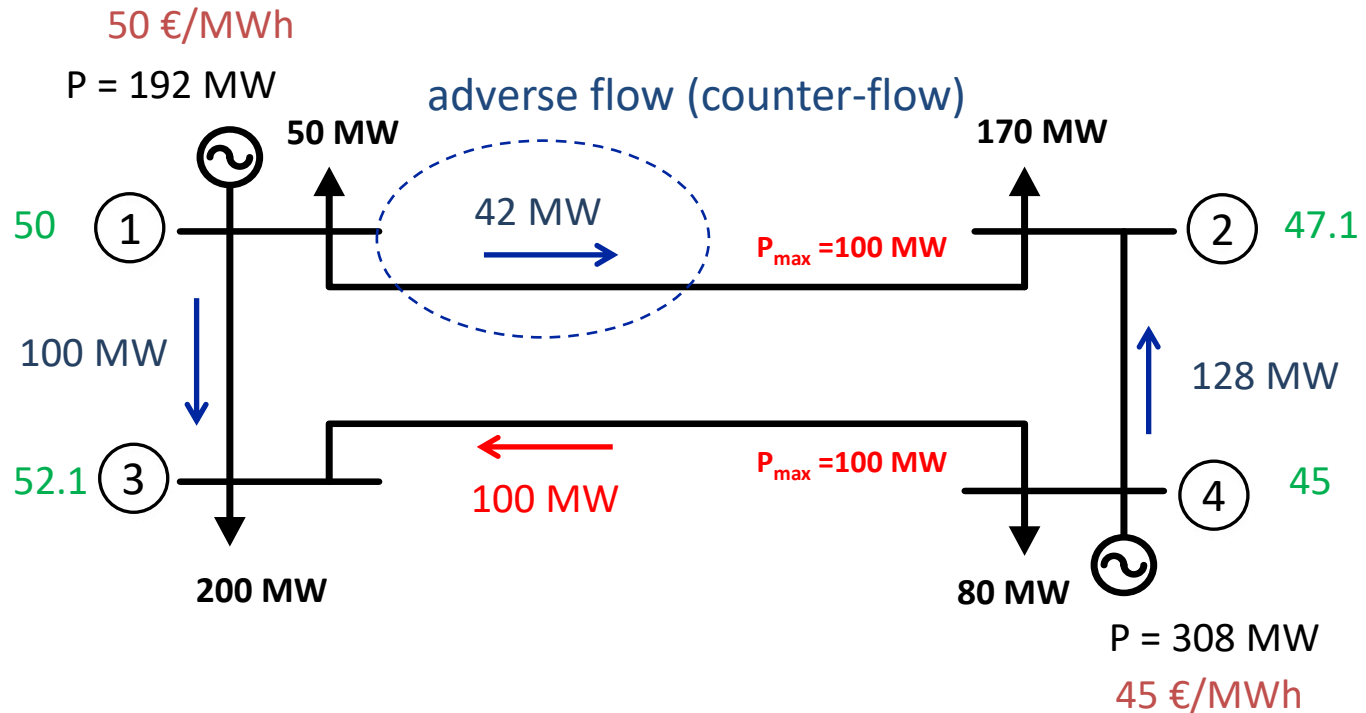
green: prices



# DC OPF model with same constraints

- Cost = 23460.2 €/hour

green: prices

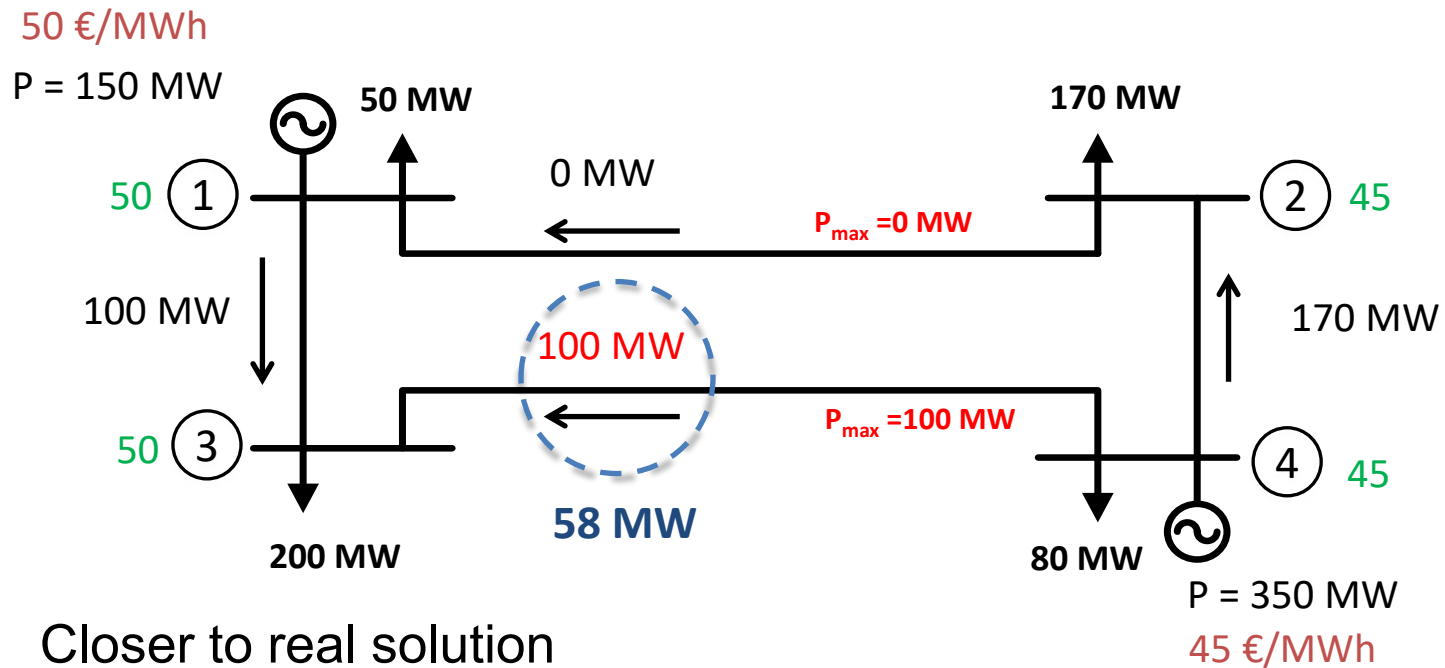


# Comments to DC OPF solution

- ATC solution cannot be obtained in reality
- Real solution more expensive
  - more constraints
  - Different prices in all nodes
- By changing grid constraints in ATC model, solutions closer to reality can be obtained
- E.g. set constraint on line 1-2 to 0 MW in ATC model
  - it appears that 42 MW gives a solution with flow in the opposite direction

# ATC model with modified constraints

green: prices



- Closer to real solution
- By setting constraint of 58 MW on the flow from 3-4 the same dispatch solution is obtained with the ATC model as with the OPF model

# Further comments

- ATC model closer to physical reality by "tuning" ATC values
  - in reality very rough because
    - uncertainty
    - same value used for long periods
- ATC "zonal model" results in two prices, 50 for node 1 and 3 and 45 for node 2 and 4
- But: the real flow in the grid is from node 1 to node 2, i.e. from high price to low price!
  - however, the total flow between the areas is from low price to high price

# Market clearing considering network

# Market clearing considering network

Let's use an **approximate linearized** representation of power flow equations (**DC power flow**). Accordingly, the following equation gives the power flow across the line connecting bus  $n$  to bus  $m$ :

$$f_{n,m} = B_{n,m} (\theta_n - \theta_m)$$

**Power flow from bus  $n$  to bus  $m$  (variable)**

**Susceptance (parameter)**

**Difference of voltage angles of buses  $n$  and  $m$  (variable)**

# Market clearing considering network

$$\text{Max}_{p_g^G, p_d^D, \theta_n} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G$$

subject to

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G$$

$$\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad (\lambda^n) \quad \forall n$$

$$-T_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq T_{n,m} \quad \forall n, \forall m$$

$$\theta_{ref} = 0$$

All demands  
located at bus  $n$

All generators  
located at bus  $n$

All buses  $m$  connected  
to bus  $n$  through  
transmission lines

Nodal price at bus  $n$   
(locational marginal price, LMP)



# Market clearing considering network

$$\text{Max}_{p_g^G, p_d^D, \theta_n} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G$$

subject to

$$0 \leq p_d^D \leq \bar{P}_d^D \quad d = 1 \dots N_D$$

$$0 \leq p_g^G \leq \bar{P}_g^G \quad g = 1 \dots N_G$$

$$\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad (\lambda^s) \quad \forall n$$

$$-T_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq T_{n,m} \quad \forall n, \forall m$$

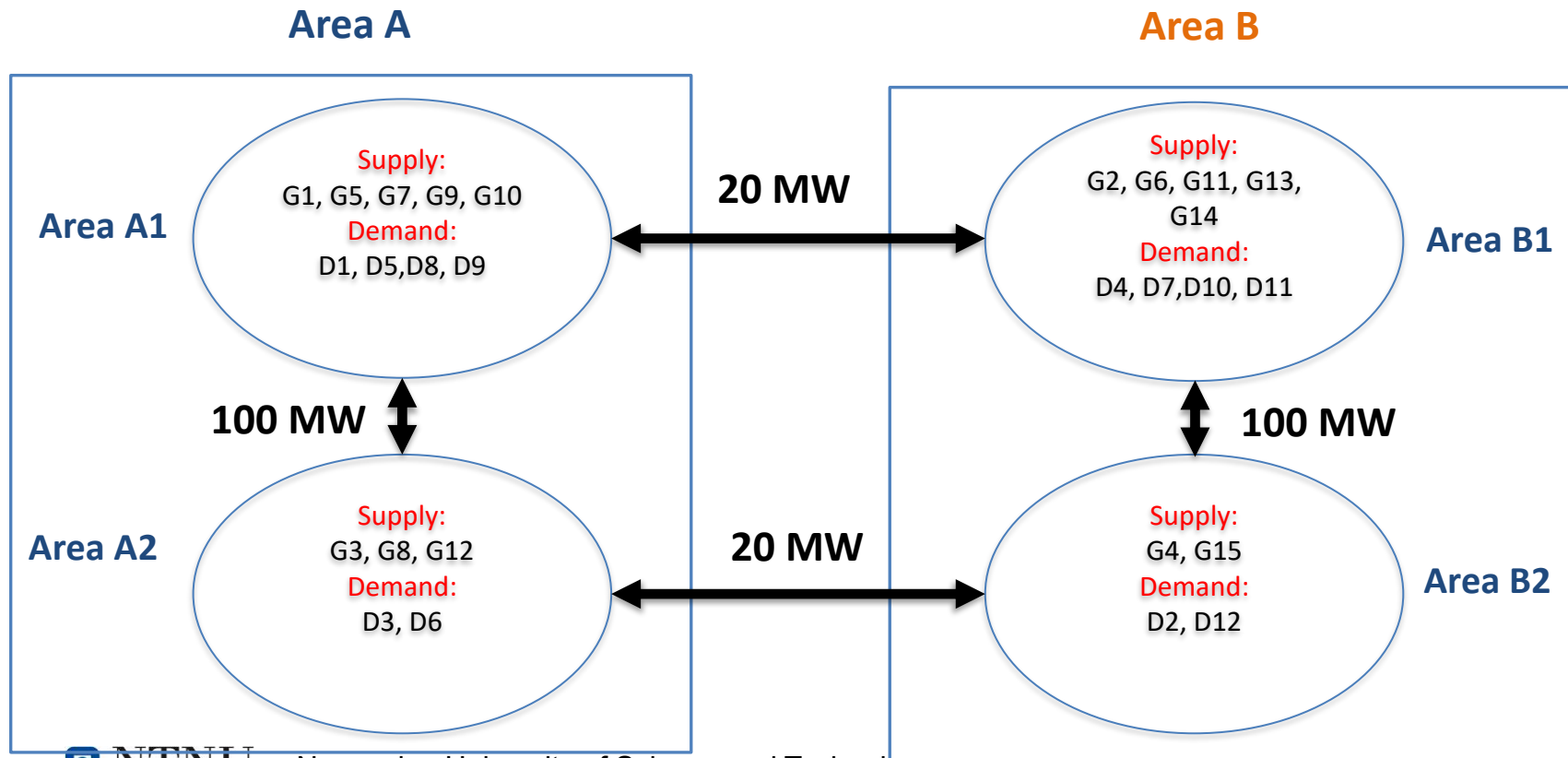
$$\theta_{ref} = 0$$

Voltage angle at the  
reference bus

Capacity of line  
connecting bus  $n$  to bus  $m$

# Market clearing considering network

- Let us revisit our previous market clearing example again,
  - considering two sub-areas within both areas **A** (A1 & A2) and **B** (B1 & B2)
  - with a transmission capacity indicated



# Localization of offers

Demand: (for a total of 1065 MWh)

| Company | id  | $P_d^D$ (MW) | $U_d$ (€/MWh) | Area      |
|---------|-----|--------------|---------------|-----------|
| R1      | D1  | 250          | 200           | <b>A1</b> |
| R2      | D2  | 300          | 110           | <b>B2</b> |
| R3      | D3  | 120          | 100           | <b>A2</b> |
| R4      | D4  | 80           | 90            | <b>B1</b> |
| R5      | D5  | 40           | 85            | <b>A1</b> |
| R6      | D6  | 70           | 75            | <b>A2</b> |
| R1      | D7  | 60           | 65            | <b>B1</b> |
| R5      | D8  | 45           | 40            | <b>A1</b> |
| R4      | D9  | 30           | 38            | <b>A1</b> |
| R5      | D10 | 35           | 31            | <b>B1</b> |
| R1      | D11 | 25           | 26            | <b>B1</b> |
| R2      | D12 | 10           | 16            | <b>B2</b> |

# And on the supply side

Supply: (for a total of 1435 MWh)

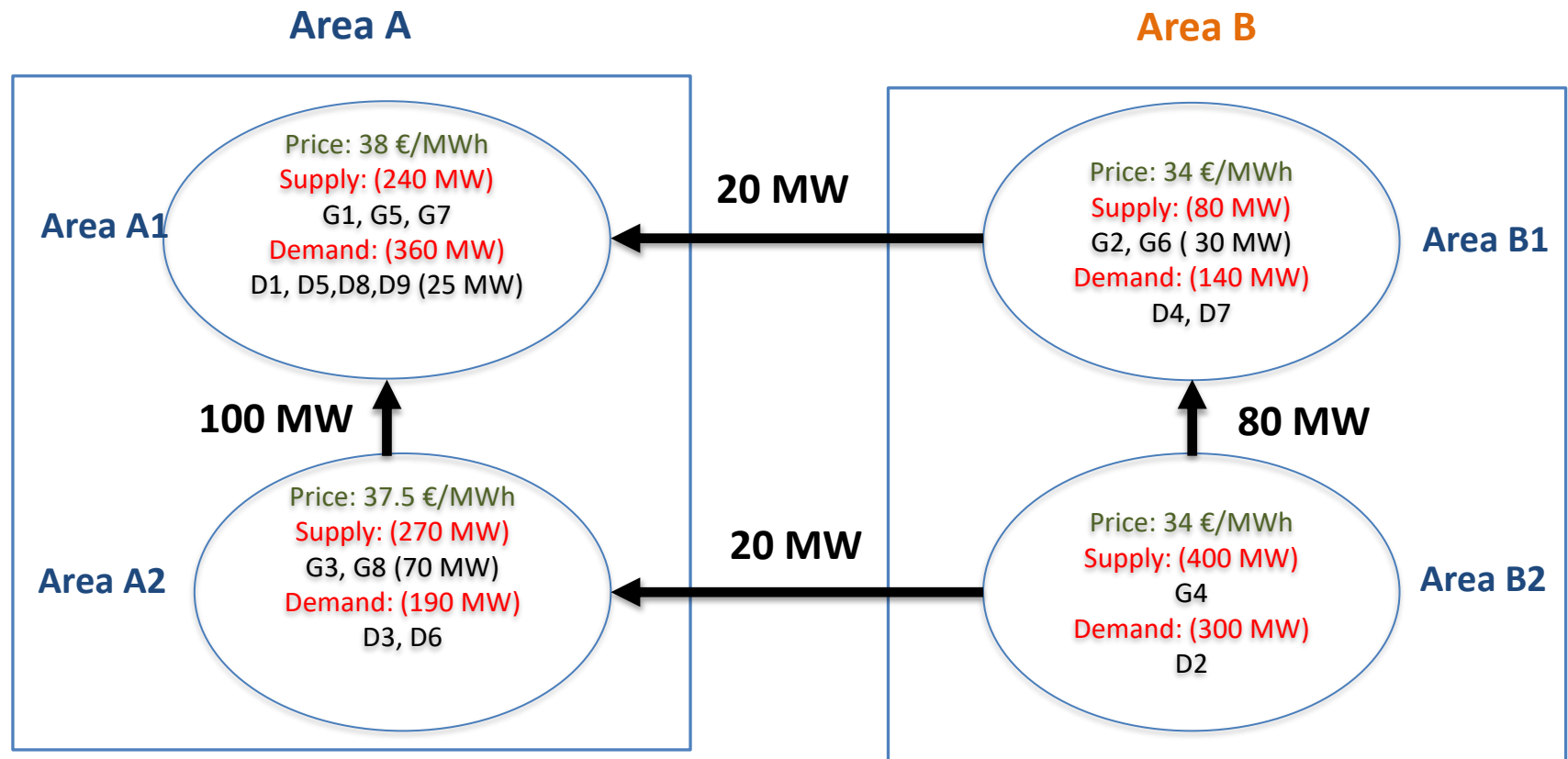
| Company | id  | $P_g^G$ (MW) | $C_g$ (€/MWh) | Area      |
|---------|-----|--------------|---------------|-----------|
| C1      | G1  | 120          | 0             | <b>A1</b> |
| C2      | G2  | 50           | 0             | <b>B1</b> |
| C3      | G3  | 200          | 15            | <b>A2</b> |
| C1      | G4  | 400          | 30            | <b>B2</b> |
| C4      | G5  | 60           | 32.5          | <b>A1</b> |
| C4      | G6  | 50           | 34            | <b>B1</b> |
| C4      | G7  | 60           | 36            | <b>A1</b> |
| C5      | G8  | 100          | 37.5          | <b>A2</b> |
| C5      | G9  | 70           | 39            | <b>A1</b> |
| C5      | G10 | 50           | 40            | <b>A1</b> |
| C1      | G11 | 70           | 60            | <b>B1</b> |
| C1      | G12 | 45           | 70            | <b>A2</b> |
| C6      | G13 | 50           | 100           | <b>B1</b> |
| C6      | G14 | 60           | 150           | <b>B1</b> |
| C6      | G15 | 50           | 200           | <b>B2</b> |

# ATC Market clearing -Accepted Bids and offers

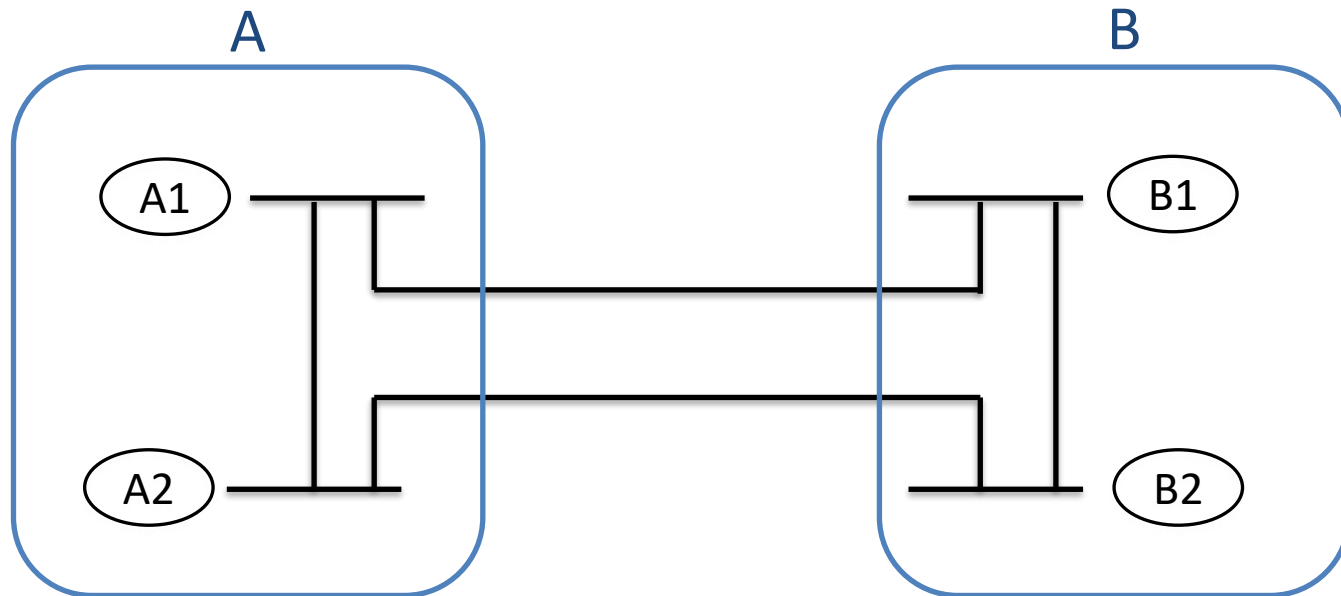
| id  | $P_d^D$ (MW) | $U_d$ (€/MWh) | Area | Total |
|-----|--------------|---------------|------|-------|
| D1  | 250          | 200           | A1   | 360   |
| D5  | 40           | 85            | A1   |       |
| D8  | 45           | 40            | A1   |       |
| D9  | 25(30)       | 38            | A1   |       |
| D3  | 120          | 100           | A2   | 190   |
| D6  | 70           | 75            | A2   |       |
| D4  | 80           | 90            | B1   | 140   |
| D7  | 60           | 65            | B1   |       |
| D10 | 35           | 31            | B1   |       |
| D11 | 25           | 26            | B1   |       |
| D2  | 300          | 110           | B2   | 300   |
| D12 | 10           | 16            | B2   |       |

| id  | $P_g^G$ (MW) | $C_g$ (€/MWh) | Area | Total |
|-----|--------------|---------------|------|-------|
| G1  | 120          | 0             | A1   | 240   |
| G5  | 60           | 32.5          | A1   |       |
| G7  | 60           | 36            | A1   |       |
| G9  | 70           | 39            | A1   |       |
| G10 | 50           | 40            | A1   |       |
| G3  | 200          | 15            | A2   | 270   |
| G8  | 70(100)      | 37.5          | A2   |       |
| G12 | 45           | 70            | A2   |       |
| G2  | 50           | 0             | B1   | 80    |
| G6  | 30(50)       | 34            | B1   |       |
| G11 | 70           | 60            | B1   |       |
| G13 | 50           | 100           | B1   |       |
| G14 | 60           | 150           | B1   | 400   |
| G4  | 400          | 30            | B2   |       |
| G15 | 50           | 200           | B2   |       |

# ATC Market clearing considering network-results



# DC OPF-Grid modelling



| line | from-to | B (per unit) | Trans cap.<br>(MW) |
|------|---------|--------------|--------------------|
| 1    | A1-A2   | 19.84        | 100                |
| 2    | A1-B1   | 26.88        | 20                 |
| 3    | A2-B2   | 26.88        | 100                |
| 4    | B1-B2   | 15.72        | 20                 |

Per unit base:  
 $S_N = 100$  MVA  
 $V_N = 230$  kV

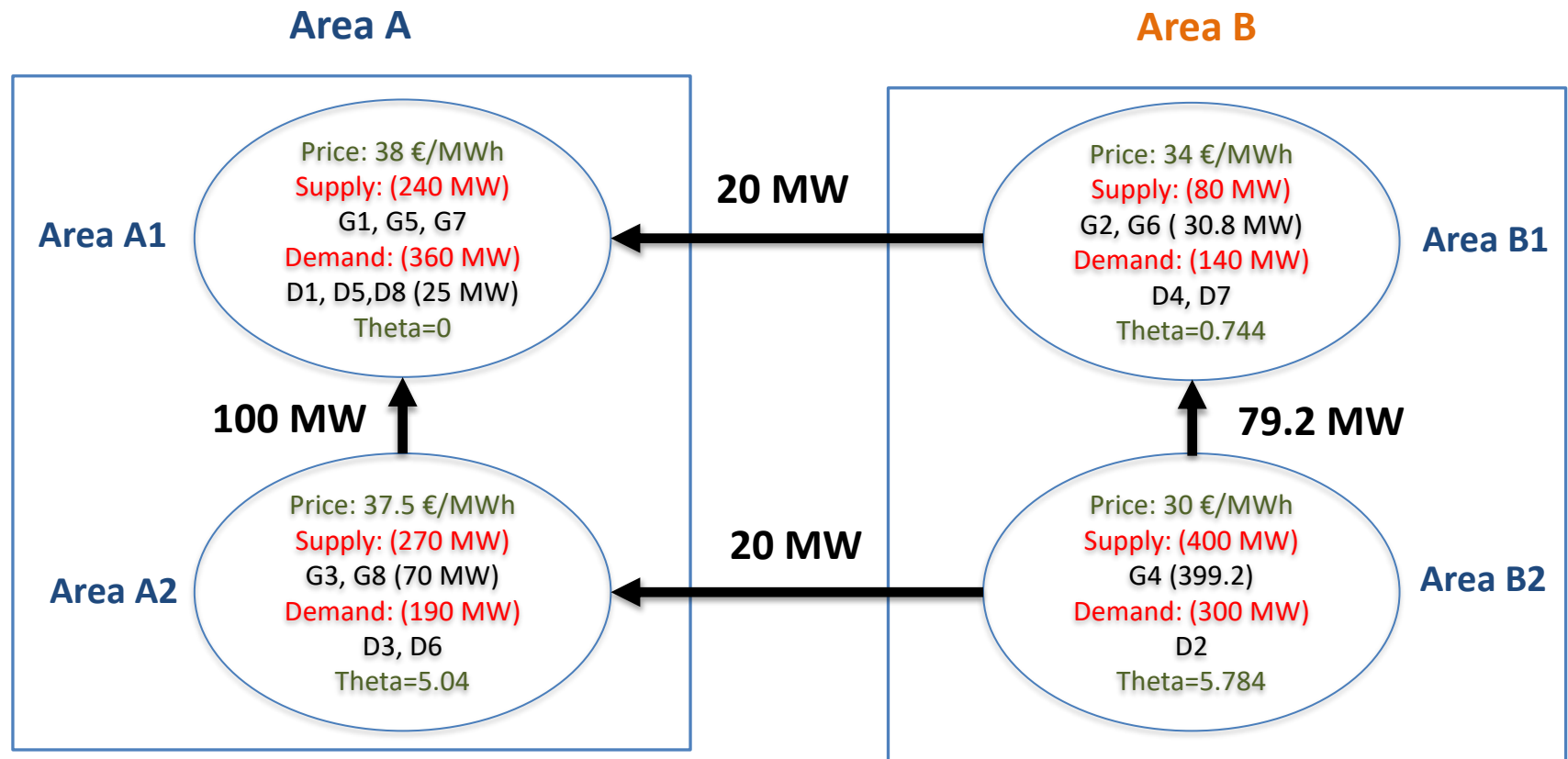
# DC OPF Market clearing -Accepted Bids and offers

| id  | $P_d^D$ (MW) | $U_d$ (€/MWh) | Area | Total |
|-----|--------------|---------------|------|-------|
| D1  | 250          | 200           | A1   | 360   |
| D5  | 40           | 85            | A1   |       |
| D8  | 45           | 40            | A1   |       |
| D9  | 25(30)       | 38            | A1   |       |
| D3  | 120          | 100           | A2   | 190   |
| D6  | 70           | 75            | A2   |       |
| D4  | 80           | 90            | B1   | 140   |
| D7  | 60           | 65            | B1   |       |
| D10 | 35           | 31            | B1   |       |
| D11 | 25           | 26            | B1   |       |
| D2  | 300          | 110           | B2   | 300   |
| D12 | 10           | 16            | B2   |       |

| id  | $P_g^G$ (MW) | $C_g$ (€/MWh) | Area | Total |
|-----|--------------|---------------|------|-------|
| G1  | 120          | 0             | A1   | 240   |
| G5  | 60           | 32.5          | A1   |       |
| G7  | 60           | 36            | A1   |       |
| G9  | 70           | 39            | A1   |       |
| G10 | 50           | 40            | A1   |       |
| G3  | 200          | 15            | A2   | 270   |
| G8  | 70(100)      | 37.5          | A2   |       |
| G12 | 45           | 70            | A2   |       |
| G2  | 50           | 0             | B1   | 80.8  |
| G6  | 30.8(50)     | 34            | B1   |       |
| G11 | 70           | 60            | B1   |       |
| G13 | 50           | 100           | B1   |       |
| G14 | 60           | 150           | B1   |       |
| G4  | 399.2 (400)  | 30            | B2   | 399.2 |
| G15 | 50           | 200           | B2   |       |



# DC OPF Market clearing considering network-results



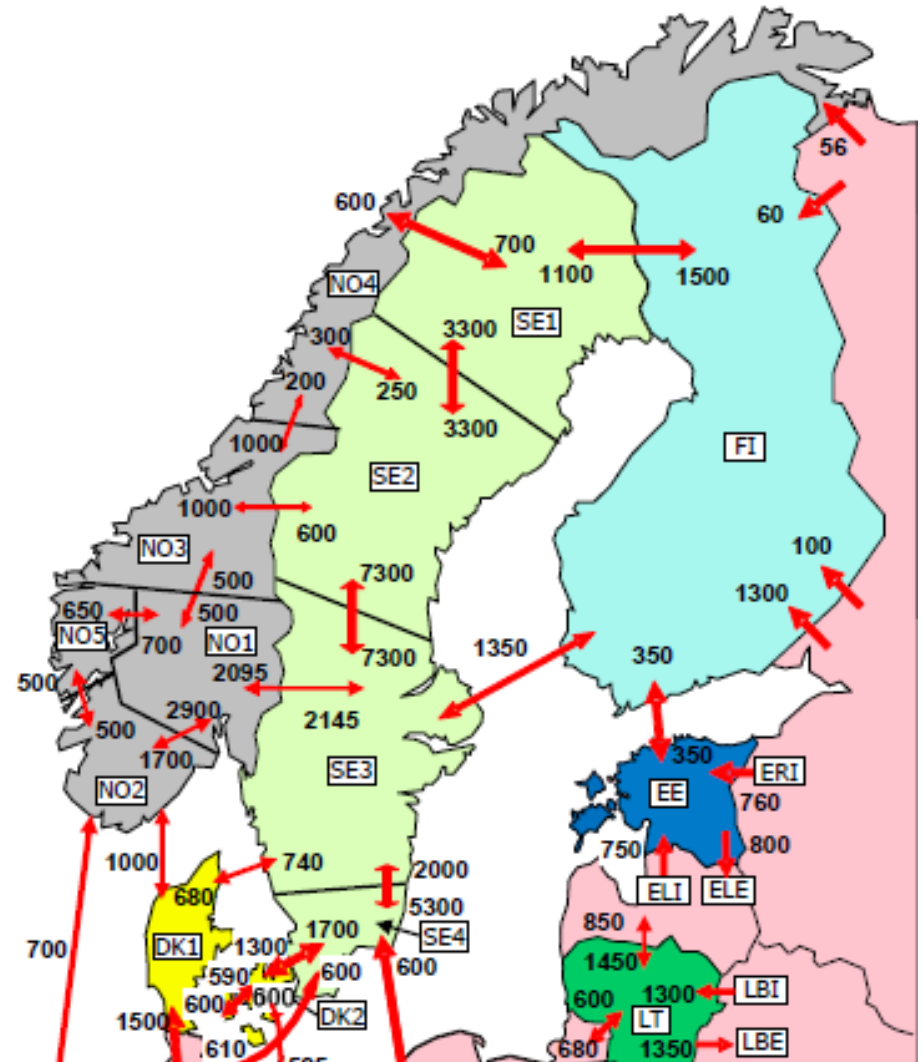
# Nordic System

Very approximate version of Nordic system to show the principles.

Use maximum transfer capacities, normally several are somewhat lower.

Approximate reactances.

Results should be interpreted as examples only

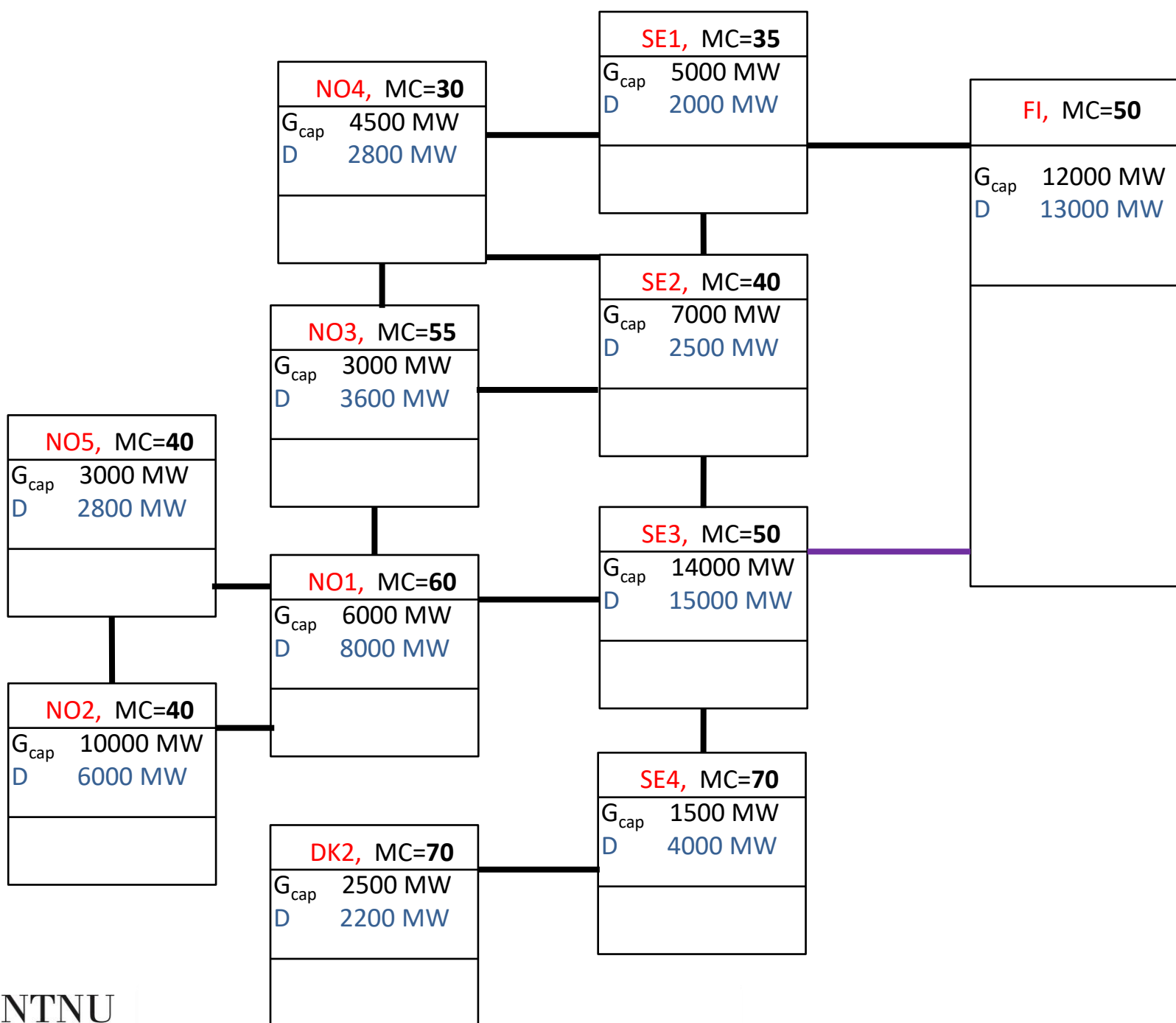


# Zone Data

|            | Pmax (MW) | Pmin (MW) | Cost (€/MWh) | Demand (MW) |
|------------|-----------|-----------|--------------|-------------|
| <b>NO5</b> | 3000      | 0         | 40           | 2800        |
| <b>NO2</b> | 10000     | 0         | 40           | 6000        |
| <b>NO1</b> | 6000      | 0         | 60           | 8000        |
| <b>NO4</b> | 4500      | 0         | 30           | 2800        |
| <b>NO3</b> | 3000      | 0         | 55           | 3600        |
| <b>SE1</b> | 5000      | 0         | 35           | 2000        |
| <b>SE2</b> | 7000      | 0         | 40           | 2500        |
| <b>SE3</b> | 14000     | 0         | 50           | 15000       |
| <b>SE4</b> | 1500      | 500       | 70           | 4500        |
| <b>FI</b>  | 12000     | 0         | 50           | 13000       |
| <b>DK2</b> | 2500      | 0         | 70           | 2200        |

# Interconnection data

| From | To  | NTC F-T (MW) | NTC T-F (MW) | Reactance (pu) |
|------|-----|--------------|--------------|----------------|
| NO4  | SE1 | 700          | 600          | 0.05754        |
| NO4  | NO3 | 1000         | 200          | 0.05111        |
| NO4  | SE2 | 250          | 300          | 0.50951        |
| NO3  | SE2 | 600          | 1000         | 0.06020        |
| NO3  | NO1 | 500          | 500          | 0.16478        |
| NO5  | NO1 | 700          | 650          | 0.08984        |
| NO5  | NO2 | 500          | 500          | 0.15676        |
| NO2  | NO1 | 2900         | 1700         | 0.03956        |
| NO1  | SE3 | 2145         | 2095         | 0.03907        |
| SE1  | SE2 | 3300         | 3300         | 0.01536        |
| SE1  | FI  | 1500         | 1100         | 0.03117        |
| SE2  | SE3 | 7300         | 7300         | 0.00842        |
| SE3  | SE4 | 5300         | 2000         | 0.00794        |
| SE4  | DK2 | 1300         | 1700         | 0.01565        |
| SE3  | FI  | 1350         | 1350         | DC             |



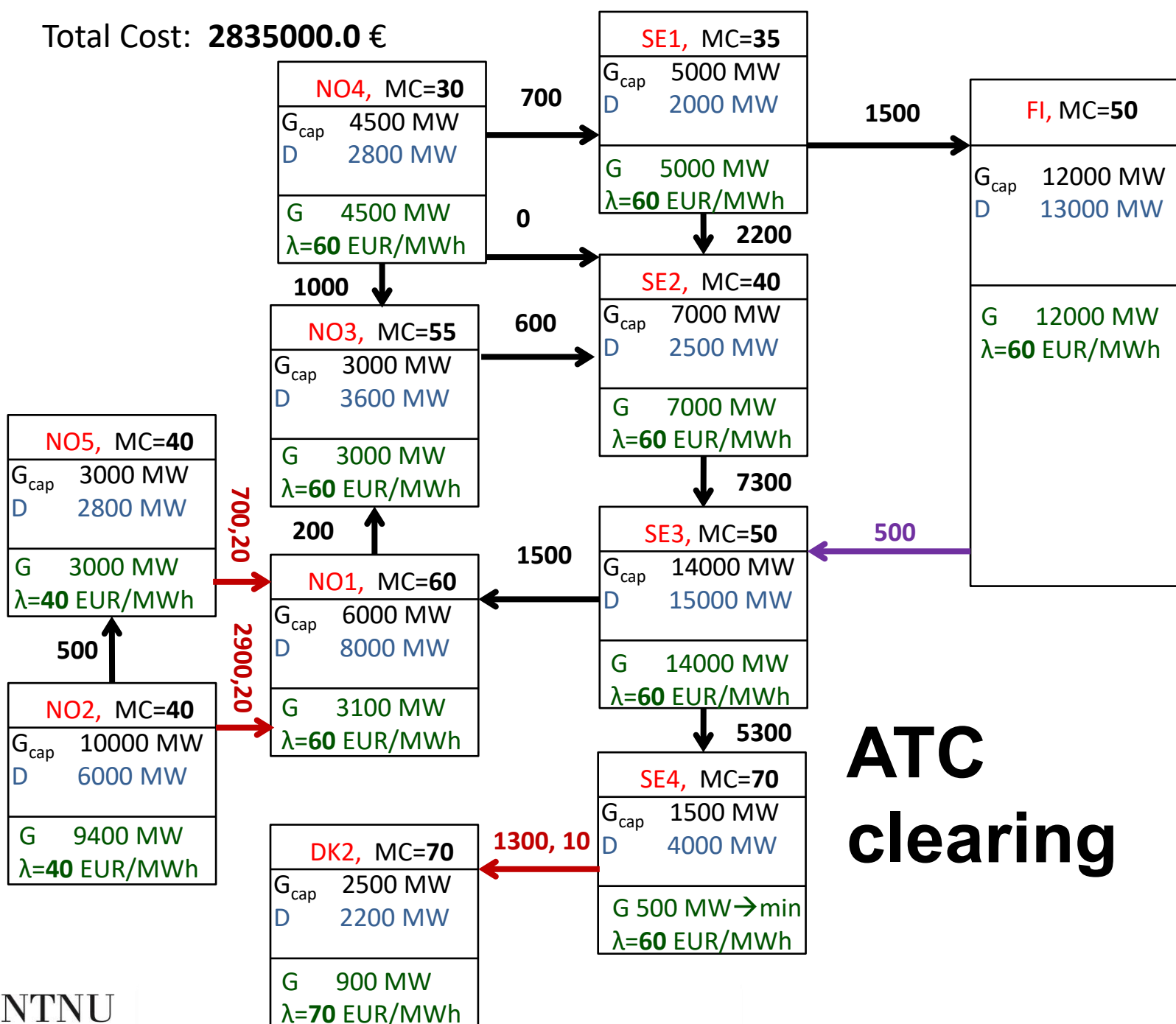
# ATC market clearing – node results

| Node | Gen<br>(MW) | maxGen<br>(MW) | Demand<br>(MW) | Price<br>(€/MWh) |          |
|------|-------------|----------------|----------------|------------------|----------|
| NO5  | 3000        | 3000           | 2800           | 40               |          |
| NO2  | 9400        | 10000          | 6000           | 40               | marginal |
| NO1  | 3100        | 6000           | 8000           | 60               | marginal |
| NO4  | 4500        | 4500           | 2800           | 60               |          |
| NO3  | 3000        | 3000           | 3600           | 60               |          |
| SE2  | 7000        | 7000           | 2500           | 60               |          |
| SE1  | 5000        | 5000           | 2000           | 60               |          |
| SE3  | 14000       | 14000          | 15000          | 60               |          |
| SE4  | 500         | 1500           | 4500           | 60               | @ MIN    |
| FI   | 12000       | 12000          | 13000          | 60               |          |
| DK2  | 900         | 2500           | 2200           | 70               | marginal |

# ATC market clearing – flow results

| From node | To node | Flow (MW) | Max flow (MW) | Dual (€/MWh) |
|-----------|---------|-----------|---------------|--------------|
| NO4       | SE1     | 700       | 700           | 0            |
| NO4       | NO3     | 1000      | 1000          | 0            |
| NO4       | SE2     | 0         | 250           | 0            |
| NO3       | SE2     | 600       | 600           | 0            |
| NO3       | NO1     | -200      | 500           | 0            |
| NO5       | NO1     | 700       | 700           | 20           |
| NO5       | NO2     | -500      | 500           | 0            |
| NO2       | NO1     | 2900      | 2900          | 20           |
| NO1       | SE3     | -1500     | 2095          | 0            |
| SE1       | SE2     | 2200      | 3300          | 0            |
| SE1       | FI      | 1500      | 1500          | 0            |
| SE2       | SE3     | 7300      | 7300          | 0            |
| SE3       | SE4     | 5300      | 5300          | 0            |
| SE4       | DK2     | 1300      | 1300          | 10           |
| SE3       | FI      | -500      | 1350          | 0            |

Total Cost: **2835000.0 €**



**ATC  
clearing**



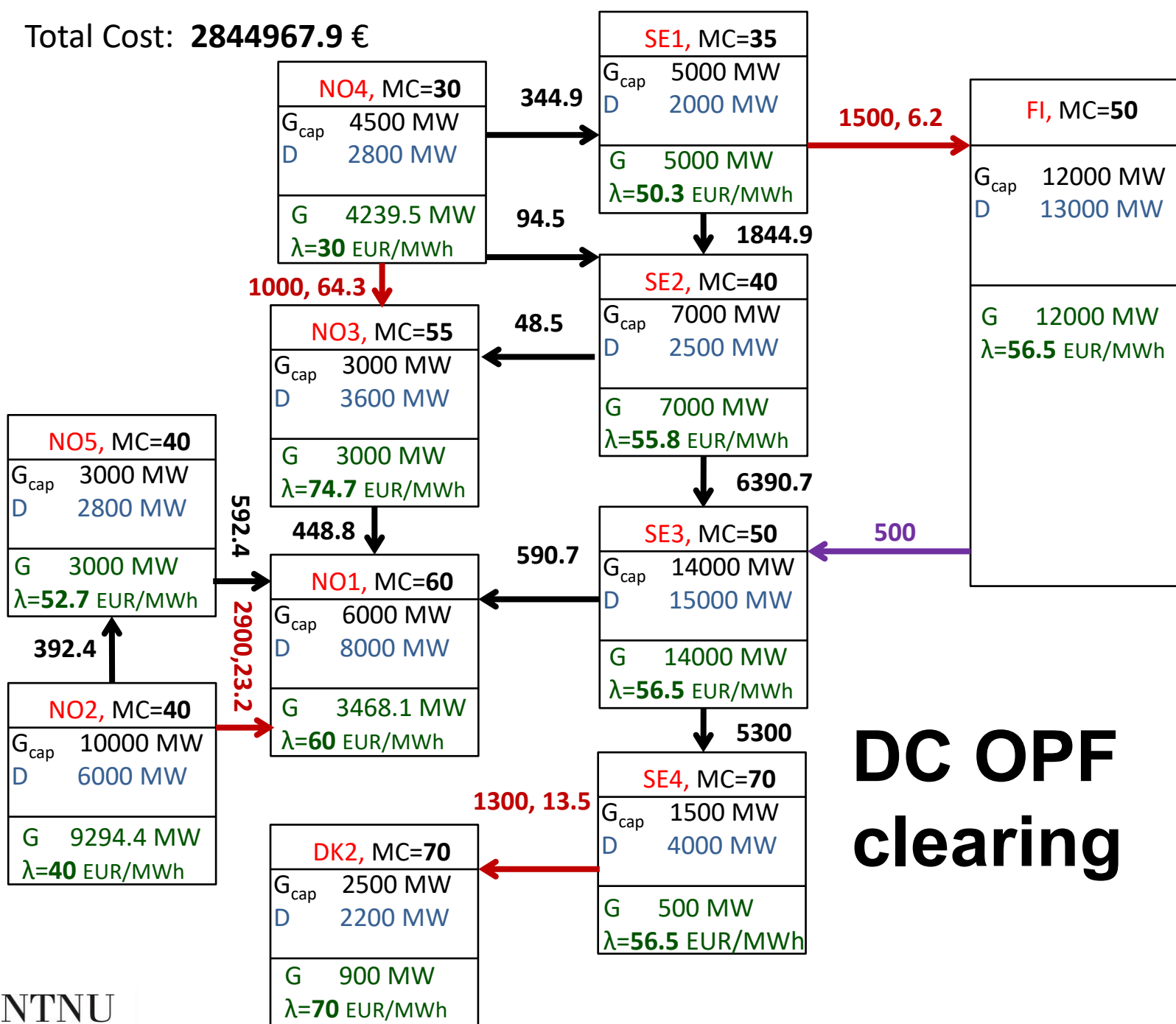
# DC OPF market clearing– node results

| Node | Gen<br>(MW) | maxGen<br>(MW) | Demand<br>(MW) | Angle<br>(rad) | Price<br>(€/MWh) |
|------|-------------|----------------|----------------|----------------|------------------|
| NO5  | 3000        | 3000           | 2800           | -0.052         | 52.7             |
| NO2  | 9292.4      | 10000          | 6000           | 0.0095         | 40.0             |
| NO1  | 3468.1      | 6000           | 8000           | -0.1052        | 60.0             |
| NO4  | 4239.5      | 4500           | 2800           | 0.0198         | 30.0             |
| NO3  | 3000        | 3000           | 3600           | -0.0313        | 74.7             |
| SE1  | 5000        | 5000           | 2000           | 0              | 50.3             |
| SE2  | 7000        | 7000           | 2500           | -0.0283        | 55.8             |
| SE3  | 14000       | 14000          | 15000          | -0.0821        | 56.5             |
| SE4  | 500         | 1500           | 4500           | -0.1242        | 56.5             |
| FI   | 12000       | 12000          | 13000          | -0.0468        | 56.5             |
| DK2  | 900         | 2500           | 2200           | -0.1446        | 70.0             |

# DC OPF market clearing– flow results

| From node | To node | Flow (MW) | Max flow (MW) | Dual (€/MWh) |
|-----------|---------|-----------|---------------|--------------|
| NO4       | SE1     | 344.9     | 700           | 0            |
| NO4       | NO3     | 1000      | 1000          | 65.4         |
| NO4       | SE2     | 94.6      | 250           | 0            |
| NO3       | SE2     | -48.8     | 1000          | 0            |
| NO3       | NO1     | 448.8     | 500           | 0            |
| NO5       | NO1     | 592.4     | 700           | 0            |
| NO5       | NO2     | -392.4    | 500           | 0            |
| NO2       | NO1     | 2900      | 2900          | 23.2         |
| NO1       | SE3     | -590.7    | 2095          | 0            |
| SE1       | SE2     | 1844.9    | 3300          | 0            |
| SE1       | FI      | 1500      | 1500          | 6.2          |
| SE2       | SE3     | 6390.7    | 7300          | 0            |
| SE3       | SE4     | 5300      | 5300          | 0            |
| SE4       | DK2     | 1300      | 1300          | 13.5         |
| SE3       | FI      | -500      | 1350          | 0            |

Total Cost: **2844967.9 €**



# ATC vs OPF

- ATC sees fewer constraints
  - more equal prices
- Need to adapt ATC values to what is possible in the real grid
- Large deviations make it necessary to correct with counter trade
- DC OPF:
  - physical characteristics of grid taken directly into account
  - better utilization of total grid resources

# Why Flow-Based Coupling

- DCOPF = full nodal physical model
- European markets clear in bidding zones
- We need:
  - Physics-aware constraints
  - Zonal price formation
  - Computational scalability
- **Core Idea:** Embed grid physics into the market without solving voltage angles in the market clearing



# Classical DCOPF Formulation

## Variables

- $P_g^G, P_d^D$ : generation and demand dispatch
- $\theta_n$ : voltage angle

$$\text{Max}_{p_g^G, p_d^D, \theta_n} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G \longrightarrow \text{Objective}$$

subject to

$$\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad (\lambda^s) \quad \forall n \longrightarrow \text{Power balance}$$

$$-T_{n,m} \leq B_{n,m} (\theta_n - \theta_m) \leq T_{n,m} \quad \forall n, \forall m \longrightarrow \text{Transmission limits}$$

$$0 \leq p_g^G \leq \bar{p}_g^G \quad g = 1 \dots N_G \longrightarrow \text{Generation limits}$$

$$0 \leq p_d^D \leq \bar{p}_d^D \quad d = 1 \dots N_D \longrightarrow \text{Demand limits}$$

$$\theta_{ref} = 0$$

- DCOPF solves physics directly
- Voltage angles are explicit decision variables

# Write line flows as a linear function of angles

- For each line  $\ell \equiv (i \rightarrow j)$  with susceptance  $b_\ell$ ,

$$f_{ij} = b_{ij} (\theta_i - \theta_j)$$

- Matrix notation of line flow:

- Let  $A \in \mathbb{R}^{L \times N}$  be the connection matrix or branch-bus incidence matrix (row  $\ell$ : +1 at “from” bus, -1 at “to” bus).
- Let  $D_b = \text{diag}(b_\ell)$ .

Then:

$$B_f = D_b A \quad \Rightarrow \quad f = D_b A \theta$$

- And the bus admittance matrix:

$$B_{bus} = A^T D_b A$$

# What is a PTDF (Power Transfer Distribution Factor) ?

- A PTDF measures: How much power flows on a transmission line when 1 MW is injected at one node and withdrawn at another (reference) node.
- Mathematical Definition:

Under the DC approximation:

$$f_k = \sum_n PTDF_{k,n} p_n$$

Where:

- $f_k$  = flow on line  $k$
- $p_n$  = net injection at node  $n$
- $PTDF_{k,n}$  = sensitivity of line  $k$  to injection at node  $n$



# Calculate PTDF from DCOPF

In FB market model, line flows are linear functions of injections. We can eliminate voltage angles

- Under DC approximation:  $\sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G + \sum_{m \in \Omega_n} B_{n,m} (\theta_n - \theta_m) = 0 \quad \forall n$
- Define the net injection at bus  $n$ :  $P_n := \sum_{d \in \Psi_n} p_d^D - \sum_{g \in \Psi_n} p_g^G$
- The node balance becomes the standard DC relation:  $P = B_{bus} \theta$  and
- Eliminate angles**: Because bus admittance matrix  $B_{bus}$  is singular, in DCOPF fixes a reference:  $\theta_{ref} = 0$
- So, we get an invertible reduced matrix:
  - $\tilde{\theta} \in \mathbb{R}^{(N-1)}$ : angles without the slack bus
  - $\tilde{p} \in \mathbb{R}^{(N-1)}$ : injections without the slack bus
  - $\tilde{B}_{bus} \in \mathbb{R}^{(N-1) \times (N-1)}$ : Reduced bus admittance matrix without the slack bus row and column
  - $\tilde{B}_f \in \mathbb{R}^{L \times (N-1)}$ : connection matrix without the slack column
- Then  $\tilde{P} = \tilde{B}_{bus} \tilde{\theta} \Rightarrow \tilde{\theta} = \tilde{B}_{bus}^{-1} \tilde{P}$
- Insert into the line-flow equation:  $f = \tilde{B}_f \tilde{\theta} = \tilde{B}_f \tilde{B}_{bus}^{-1} \tilde{P}$

**PTDF**

$$f = \tilde{B}_f \tilde{\theta} = \tilde{B}_f \tilde{B}_{bus}^{-1} \tilde{P}$$

# PTDF matrix

- The PTDF mapping from (reduced) nodal injections to line flows is:

$$H = \tilde{B}_f \tilde{B}_{bus}^{-1} \Rightarrow f = H\tilde{P}$$

- $H_{l,n}$  is the change in flow on line  $l$  caused by a +1 MW injection at bus  $n$  and a corresponding -1 MW withdrawal at the slack bus.
- $H \in \mathbb{R}^{L \times (N-1)}$ , where  $L$ =number of lines and  $N$ =number of nodes.  
Therefore, Each element:  $H_{k,n} = PTDF_{k,n}$

- The line constraints in DCOPF are:

$$-T_{n,m} \leq B_{n,m}(\theta_n - \theta_m) \leq T_{n,m}$$

- After eliminating  $\theta$ , they become PTDF flow constraints:

$$-T_l \leq (H\tilde{P})_l \leq T_l$$

# PTDF matrix

- Using PTDF matrix:

$$f = Hp$$

- This is extremely powerful because:
  - No voltage angles
  - No solving Kirchhoff equations in optimization
  - Linear mapping from injections  $\rightarrow$  flows

# Flow-Based Market Model

Decision variables

- $p_g^G, p_d^D$ : generation and demand dispatch
- $NP_z$ : Net position of bidding zone  $z$

$$\text{Max}_{p_g^G, p_d^D, NP_z} SW = \sum_{d=1}^{N_D} U_d p_d^D - \sum_{g=1}^{N_G} C_g p_g^G \longrightarrow \text{Objective}$$

subject to

$$0 \leq p_d^D \leq \bar{p}_d^D \quad d = 1 \dots N_D \longrightarrow \text{Generation limits}$$

$$0 \leq p_g^G \leq \bar{p}_g^G \quad g = 1 \dots N_G \longrightarrow \text{Demand limits}$$

$$NP_z = \sum_{g=1}^{N_G} p_g^G - \sum_{d=1}^{N_D} p_d^D \quad (\lambda_z) \quad \forall z \in Z \longrightarrow \text{Zonal balance}$$

$$\sum_{z=1}^Z NP_z = 0 \quad (\lambda_s) \longrightarrow \text{Global balance}$$

$$-RAM_k \leq \sum_{z=1}^Z PTDF_{k,z} NP_z \leq RAM_k \quad (\mu_k) \quad \forall k \in K \longrightarrow \text{Transmission limits}$$

$z \in Z$ : bidding zones      &       $k \in K$ : critical network elements (CNECs / monitored lines)

# Lagrangian of the Flow-Based Problem

Take the simplified welfare maximization:

max SW  
subject to:  
Zonal balance  
Global balance  
Transmission limits (PTDF  $\times$  NP  $\leq$  RAM)

$$\mathcal{L} = SW + \sum_z \lambda_z \left( NP_z - \sum_{i \in z} (g_i - d_i) \right) + \lambda_s \left( 0 - \sum_z NP_z \right) + \sum_k \mu_k \left( RAM_k - \sum_z PTDF_{k,z} NP_z \right)$$

First-Order Condition for  $NP_z$

$$\frac{\partial \mathcal{L}}{\partial NP_z} = \lambda_z - \lambda_s - \sum_k \mu_k PTDF_{k,z} = 0 \quad \Rightarrow \quad \lambda_z = \lambda_s + \sum_k \mu_k PTDF_{k,z}$$

Economic Interpretation:

Zonal price = System price + weighted sum of congestion prices  
Weights = PTDFs.

# Economic Interpretation

PTDF translates:

- Injection decisions to
  - Physical congestion
  - Market prices
- If line  $k$  is congested, the shadow price of:

$$\sum_n PTDF_{k,n} p_n \leq RAM_k$$

affects nodal or zonal prices.

# Price Formation Principle

Price in each zone =

**System Price + Congestion Component**

Where congestion component depends on:

- Binding transmission constraints
- Shadow prices of capacity constraints

If no constraint binds:

$$\mu_k = 0 \Rightarrow \text{uniform price}$$

If constraints bind:

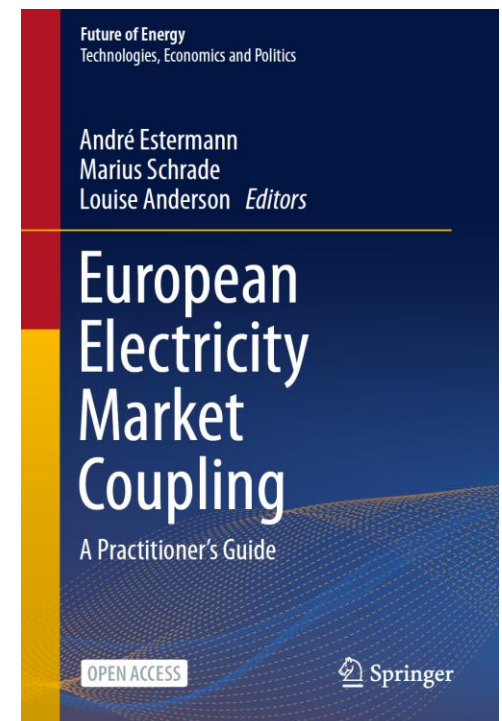
- Zonal price separation occurs
- Scarcity rents emerge

# Why Market Coupling?

## The Objective of European Market Integration

- According to the Third Energy Package (2009) and Capacity Allocation and Congestion Management (CACM) Guideline:
  - Create a single internal electricity market
  - Allocate cross-zonal capacity efficiently
  - Maximize European social welfare
  - Ensure non-discriminatory access to networks

<https://link.springer.com/book/10.1007/978-3-031-86315-8>





# What is Market Coupling

Definition (Capacity Allocation and Congestion Management (CACM) Guideline<sup>1</sup>)

Market Coupling is: A mechanism that simultaneously:

- Clears bids across multiple bidding zones
- Allocates cross-zonal transmission capacity
- Maximizes total economic surplus

Two Timeframes Covered by CACM

- Single Day-Ahead Coupling (SDAC)
- Single Intraday Coupling (SIDC)

<sup>1</sup><https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32015R1222>

# Institutional Setup-Who Does What?

## TSOs

- Calculate cross-zonal capacity
- Ensure operational security
- Receive congestion income

## Nominated Electricity Market Operators (NEMOs)

- Operate SDAC & SIDC
- Run market coupling algorithm (EUPHEMIA<sup>1</sup>)
- Nominate physical flows

## Regional Coordination Centres (RCCs)

- Support coordinated capacity calculation (regional Common Grid Model, RAM, Identification of CNEs, ...)

# What is Single Day-Ahead Coupling (SDAC)?

SDAC is: A single pan-European auction clearing day-ahead electricity for all coupled zones simultaneously

- **Timing**

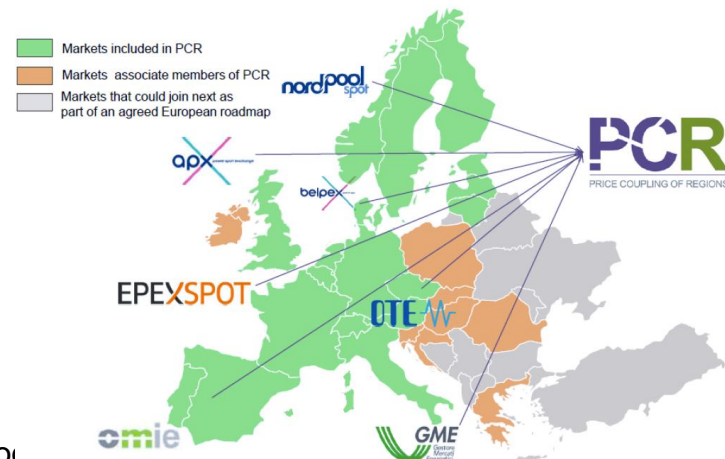
- Gate closure: 12:00 CET (D-1)
- 24 hourly prices determined
- One price per zone per Market Time Unit (MTU)

# Price Coupling of Regions (PCR)

- **WHAT is PCR?** Price Coupling of Regions (PCR) is a project of European Power Exchanges to harmonise the European electricity markets
- **HOW is this done?**
  - The common algorithm enables the determination of day-ahead electricity prices and the corresponding electricity flows between the bidding areas in a fair and transparent manner across Europe.
  - These prices reflect the cost of meeting the last unit of demand in each region, considering the availability of cheaper or more expensive generation in neighboring regions.
  - **Euphemia** is used to calculate electricity prices across Europe. It also optimises the overall welfare and increases the transparency of prices and flows

# Single Day-Ahead Coupling (SDAC)

- The aim of Single Day-ahead Coupling (SDAC) is to create a single pan European cross zonal day-ahead electricity market.
- SDAC is a cooperative effort among Nominated Electricity Market Operators (NEMOs) and Transmission System Operators (TSOs). This collaboration is structured around the Capacity Allocation and Congestion Management (CACM) guidelines, which govern how electricity transmission capacity and congestion are managed.
- SDAC uses the PCR (Price Coupling of Regions) mechanism to calculate electricity prices across different regions, integrating bids and offers from multiple markets to determine the optimal flow of electricity and corresponding prices.

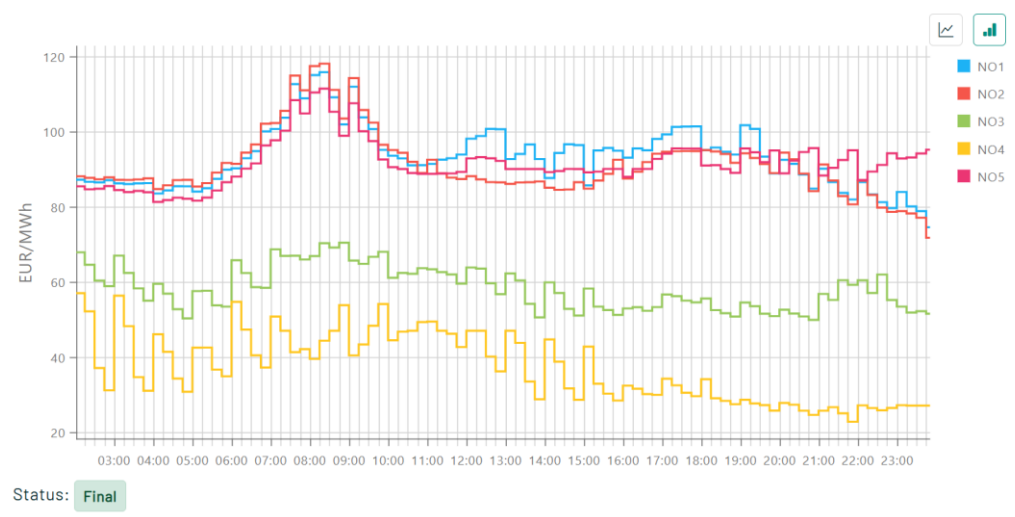


# How SDAC Works ?

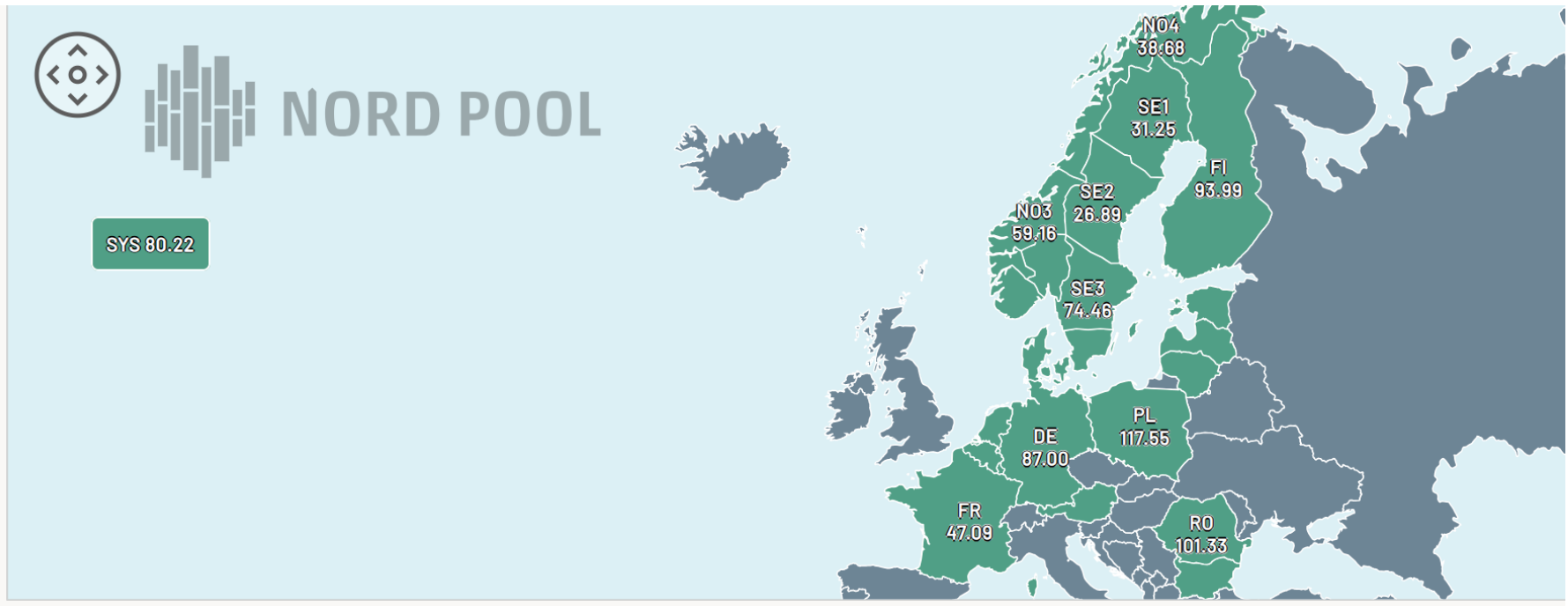
1. TSOs calculate cross-zonal capacitance
2. NEMOs collect bids
3. EUPHEMIA solves welfare maximisation.
4. Prices & flows determined simultaneously
5. Congestion income transferred to TSOs

# SDAC: Current Developments

1. 15-min Market Time Unit (MTU) in SDAC (from 2025)
2. Expansion of Flow-Based regions
3. Advanced Hybrid Coupling: Integrating HVDC interconnectors properly inside a Flow-Based market clearing framework
4. CCR mergers: Two or more Capacity Calculation Regions are merged into one larger region for capacity calculation



# Current Status of Nord Pool





# Flow-Based in Europe

FB in Europe

**Nordic: FB MC**

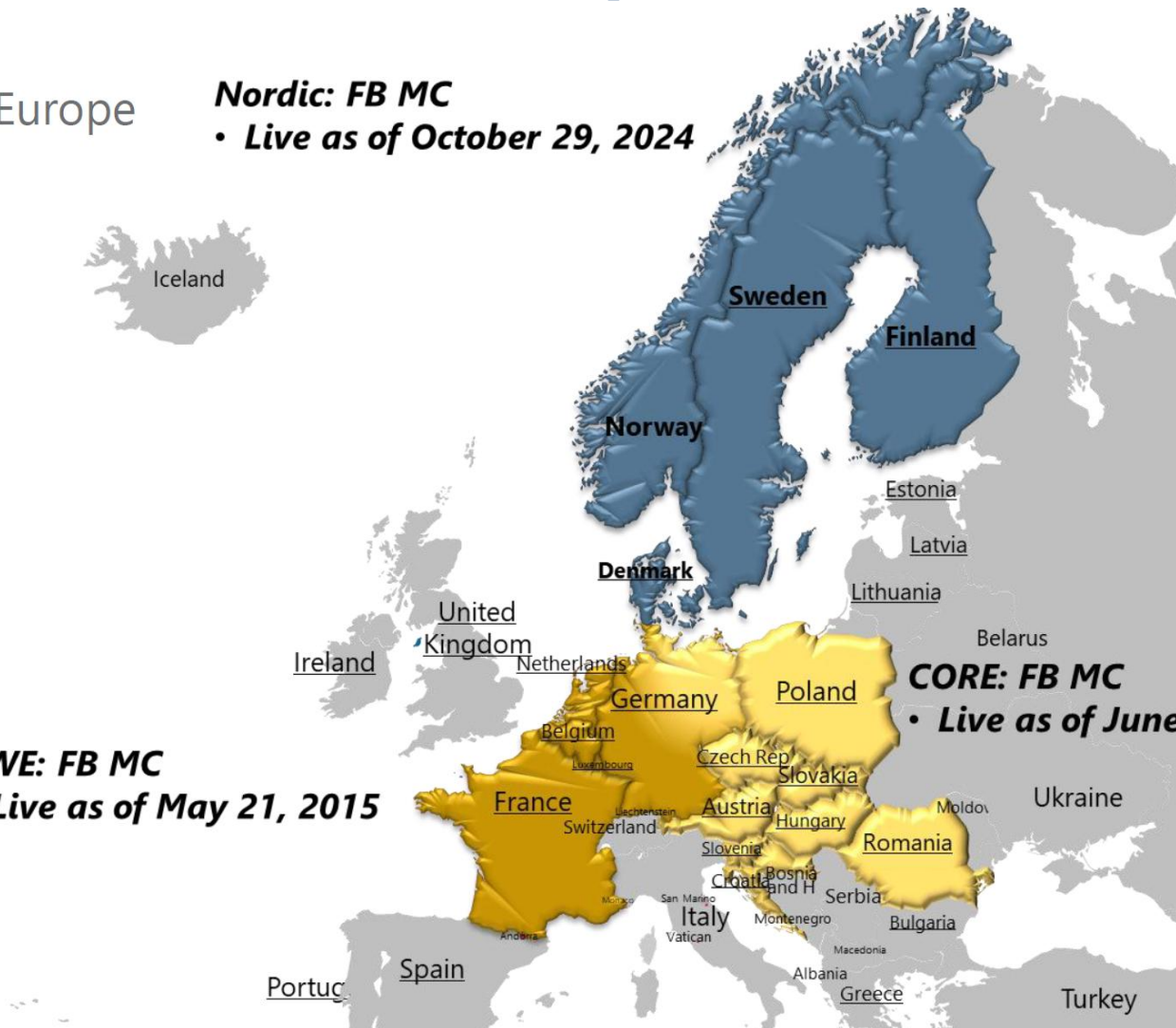
• **Live as of October 29, 2024**

**CWE: FB MC**

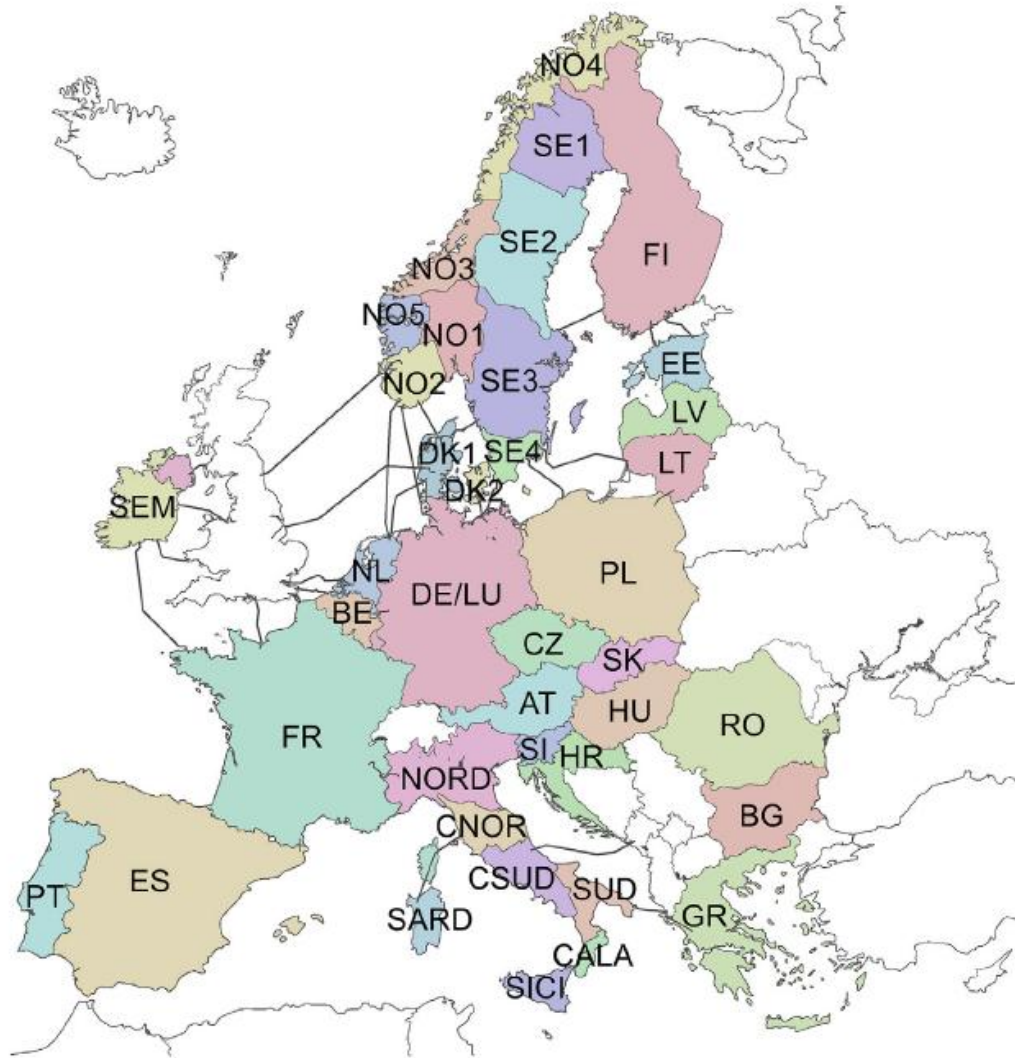
• **Live as of May 21, 2015**

**CORE: FB MC**

• **Live as of June 2022**



# Current bidding zones in Europe



# Final remarks

- **The Grid Is Not a Copper Plate:** Markets clear subject to physics, not independently of it.
- **From ATC to Flow-Based:**
  - ATC → simplified commercial limits
  - DC-OPF → full physical representation
  - PTDF → scalable mapping from injections to flows
  - Flow-Based → physics embedded in zonal markets
- **Prices = Physics + Scarcity**
  - If no constraint binds → uniform price
  - If constraints bind → congestion rents & zonal divergence
- **European Perspective:** Efficient market coupling requires economic optimization + coordinated capacity calculation